

Contextual Football Scouting

Ranking players by what they do with space, by value rather than volume

Matteo Vezzoli & Armando Mio

Barça Innovation Hub · 2 July 2026

ACT 1

Problem & positioning

Defining the research problem, framework, and contribution

The problem: team bias

- ▶ **Volume lies:** A midfielder in a possession-dominant team piles up easy passes under little pressure, and raw counts reward them all the same
- ▶ Most of a **player's value** happens **off the ball**: exactly the part the event log cannot see
- ▶ Team bias has **two parts**:
 - How each action is valued
 - How many chances a strong team hands a player

We tackle the first directly: the second, the extra chances a strong side gives, is harder to remove, and stays a declared limit

- ▶ **Our answer: grade** actions **by value**, not count, and compare players only within their role
- ▶ The effect **is measurable**: on a contextual value ranking, over half the 272-player pool shifts by 20+ percentile points against a volume proxy

Thesis and the four studies

► Read a **player's quality** from what he does with the space around him: **by value**, not volume, within macro-role

Study	What it measures	Headline metric
H1 • Space Control	Quantify a player’s spatial influence from each freeze frame, and verify the resulting indices rank players differently from volume-based statistics	Progression, Dangerousness, Reception, Gravity
H2 • Decision Quality	Grade the value of the pass a player chooses against the in-frame alternatives he ignored, and confirm the resulting index tracks the value actually created within role	DQ index
H3 • Off-Ball Movement	Measure the dangerous off-ball space a player offers but does not receive , which first requires resolving anonymous off-ball identities at validated accuracy, and test that the score is internally reliable	URS / 90
H4 • Player Similarity	Match players within role by their contextual spatial DNA on a fixed 0–100 scale, and check that the matches read as style rather than team or position	Similarity score (0–100)

Where this work sits?

- ▶ **Same conceptual family as established possession-value models:** xT/VAEP, pitch control, EPV (*Fernández–Bornn*): but ours runs entirely on a single public 360 freeze frame, trading their continuous tracking for openly available data
- ▶ **Three things set this work apart:** grading by EPV instead of just counting: every value read from open 360 data, not proprietary tracking: and judging the action against what was ignored (H2) or never served (H3), which standard models don't do
- ▶ *Fernández and Bornn* **left potential space**, the dangerous space a player could reach but doesn't yet occupy, as an open direction
- ▶ *Anzer and Bauer* model pass difficulty from tracking data: the same quantity our **xPass** model estimates from a single frame
- ▶ *Dick, Link and Brefeld* **model player availability**: close to both our **H3 receiver resolver** and the **H2 in-frame option set**

Established literature	Contextual Football Scouting
xT · VAEP · pitch control EPV (Fernández–Bornn)	xEPV single 360 frame, not tracking
Fernández & Bornn	H3 · off-ball movement
Anzer & Bauer	xPass
Dick, Link & Brefeld	H3 resolver + H2 option set

The data: what we built from, and what we built ourselves

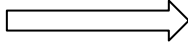
StatsBomb 360

The source: 51 matches, UEFA Euro 2024. We used a **static snapshot** of every player at the instant of each event

Filter: >135 min played

Excluded from the total:

- -28 goalkeepers
- -57 outfield players (<135 min)

357 (total)  **272** (filtered)
Analyzed Players Analyzed Players

Coverage and completion: the raw picture

Frame coverage:

- median: 87.5%
- range: 62.6%–98.2%

Raw pass completion rate:

- median: 83%
- range: 51%–97%

StatsBomb ID
↔
Transfermarkt Name

Transfermarkt

What we scraped: A standalone pipeline



Existing rows are **never duplicated**, and only missing values are filled on re-run

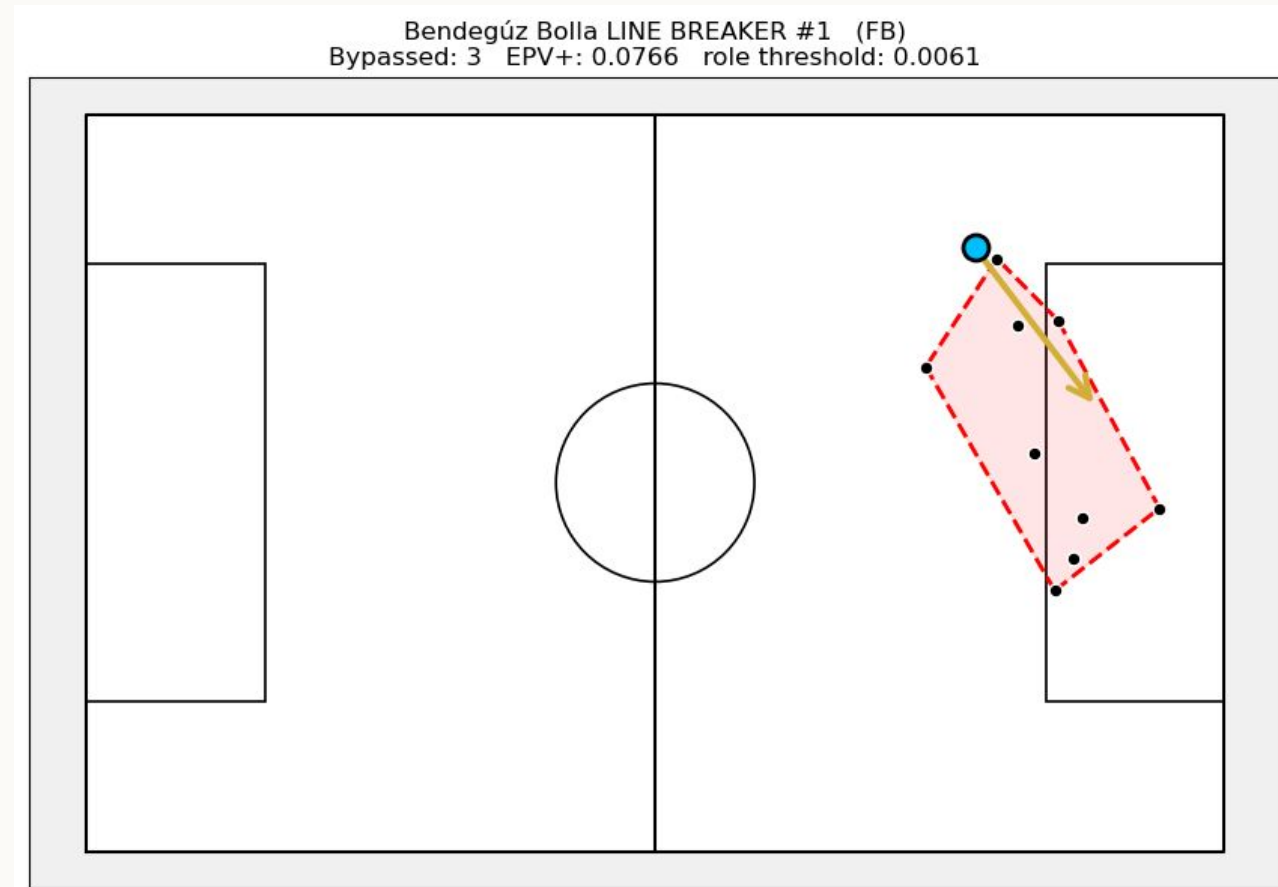
ACT 2

The shared machine

One geometry, one currency, two models we built and tested ourselves

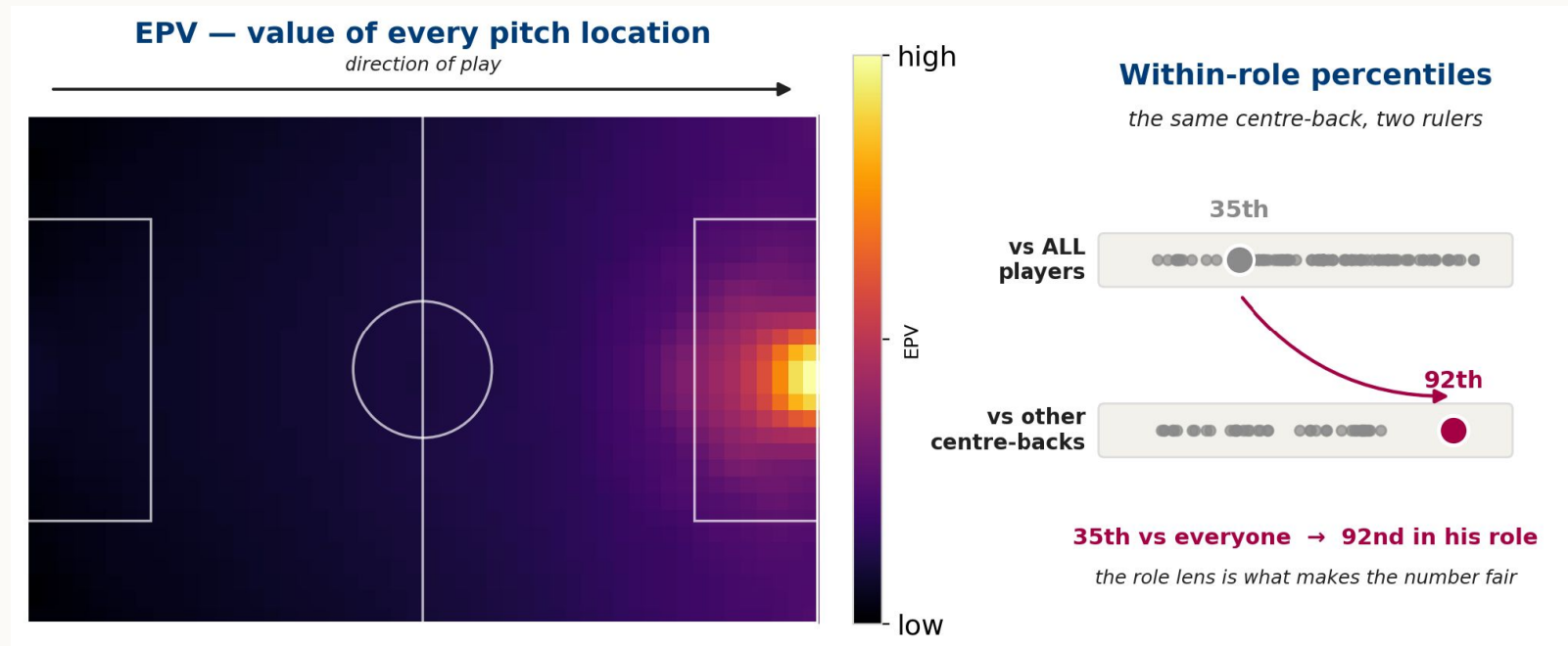
The defensive block as a convex hull

- ▶ From each freeze frame we take the visible opponents and draw their **convex hull**: the smallest polygon that encloses them all. That polygon is our working model of the **defensive block**
- ▶ We chose it because it is the **simplest object** that answers the one question a scout cares about: is the ball inside or outside the shape, and does this action move it across the boundary?
- ▶ It gives every action a **clean type**: a pass that ends inside has entered the block, one that starts inside and ends outside has exited it, a touch taken inside is a **between-the-lines reception**
- ▶ Line-breaks are then **weighted by EPV**, not just counted, so a frequent cheap line-breaker is told apart from one whose every break is dangerous



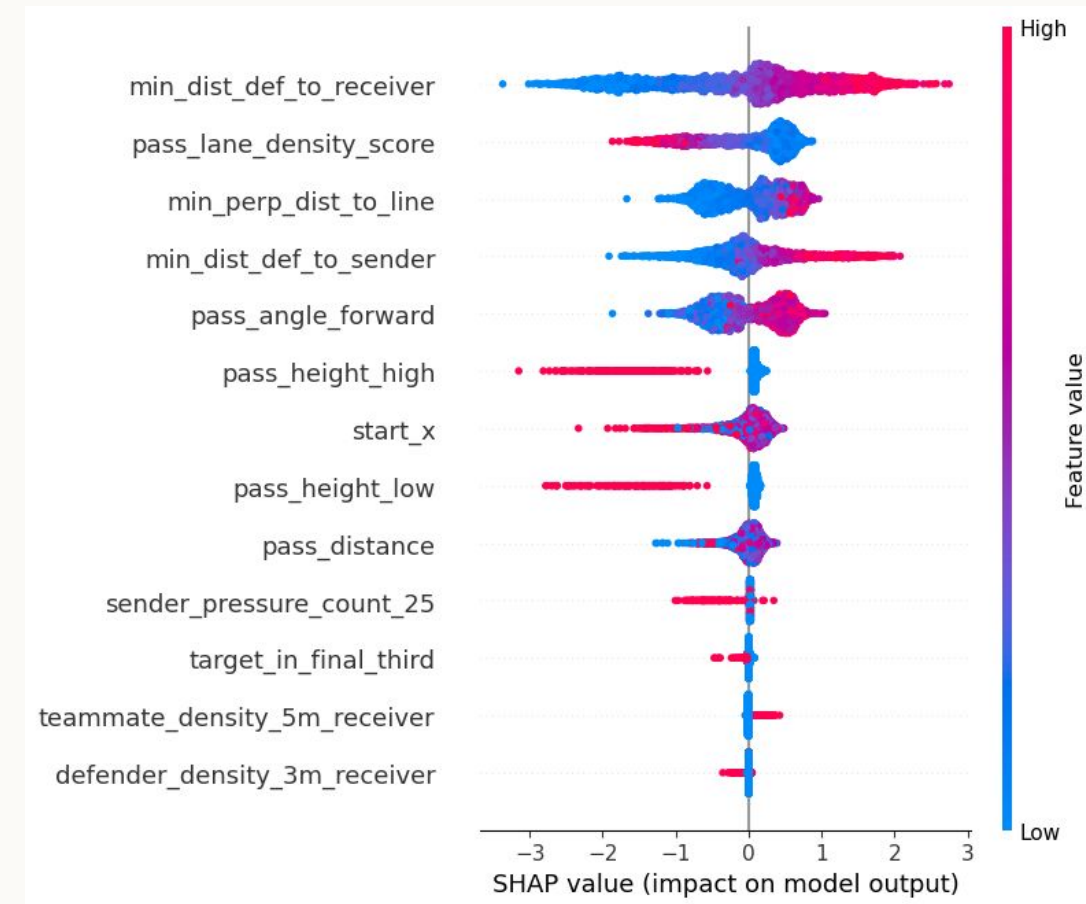
EPV and within-role: the two unifying choices

- **EPV**, the common currency: the **probability of scoring** in the next actions given the ball location. Credit only for **making the ball more dangerous**, so volume in a dominant side buys nothing. Grid adapted from *Friends-of-Tracking-Data*
- **Within-role percentiles**, the **common lens**: a centre-back is ranked only against centre-backs. **Role conditions everything** (touches inside the block: ~4% for CBs vs ~52% for CAMs), so a pitch-wide ranking would just rank roles



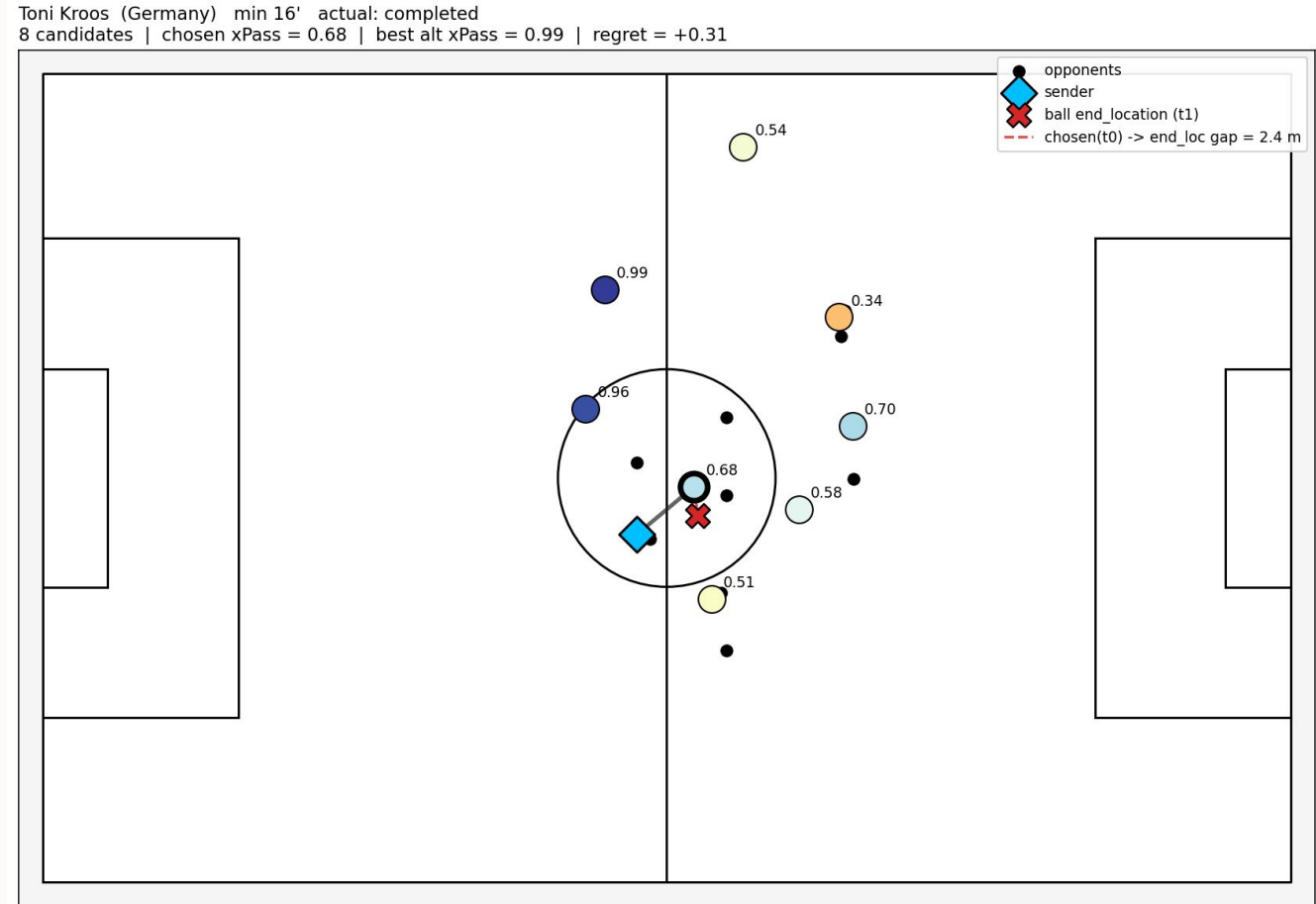
xPass: our pass-completion model

- **Our answer to will this pass arrive?** We estimated from a single 360 frame a **gradient-boosted model** trained on 30,913 real open-play passes (27,901 completed, 3,012 failed)
- **12 features** read off the frame at the **moment of release**: pass distance, forward angle, nearest-defender distance to the receiver, pass-lane congestion, opponents pressing the sender
- **Anti-leakage by design**: every feature is read at release, never from the outcome, and **cross-validation** is grouped by match so the model can't memorise a match and report a leaky score
- Chosen on the numbers: the gradient-boosted model beat L2-logistic and a balanced variant on all three metrics (**AUC 0.923, log-loss 0.190, Brier 0.057**), so it is both the best ranker and the best calibrated before any post-hoc step
- **We need probabilities, not just a ranking**, because xEPV multiplies xPass: a 0.9 that should be 0.7 would distort every score. So **we calibrate** (sigmoid, on an honest 50/50 match holdout) and clip to [0.05, 0.99]



xPass on a real frame

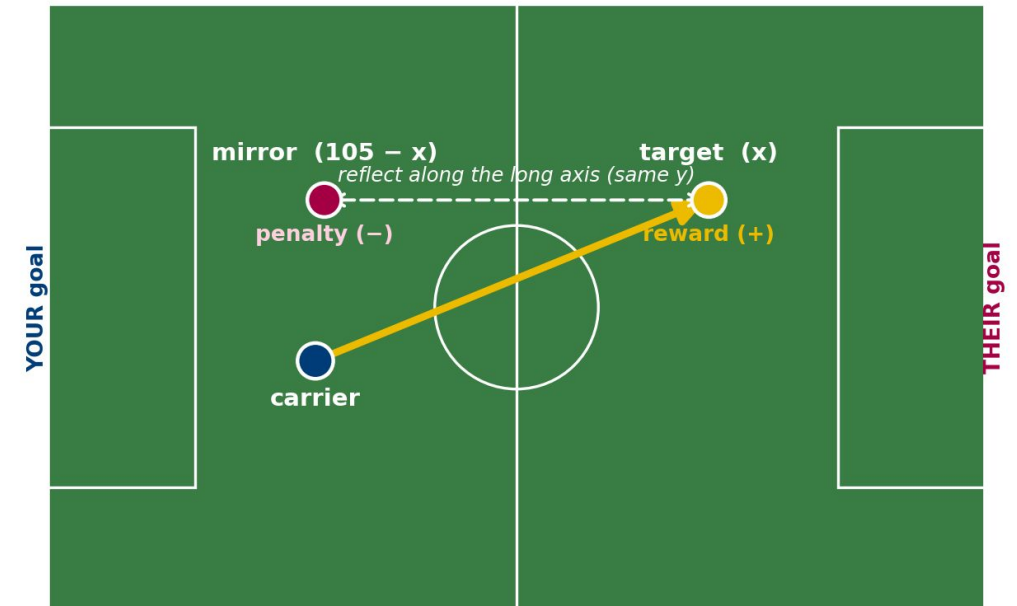
- One frame, every teammate scored for 'will this pass arrive?'. **Kroos vs Scotland**, minute 16: eight in-frame candidates, coloured by **completion probability** (red low, blue high)
- The model reads the geometry the way intuition does: the **open short ball** sits near **0.99**, a **tighter lane** through traffic drops toward **0.5**, and a long ball into a crowd is lower still
- *Kroos* chose the candidate at **xPass 0.68**, only the 4th most likely of the eight, while a near-certain 0.99 option stood open. On completion probability alone, that looks like the wrong pick
- This per-frame score is the **raw data that H2 converts into 'Decision Quality'**



xEPV: our metric, and its one elegant idea

- ▶ **xEPV is the expected net value** of attempting a pass. Two terms on one pass: $\text{xEPV} = \text{xPass} \cdot \text{EPV}(x) - (1 - \text{xPass}) \cdot \text{EPV}(105 - x)$
- ▶ **Term 1, the reward:** what the ball would be worth if the pass arrives, weighted by how likely it is to arrive
- ▶ **Term 2, the mirrored cost:** if you lose it, the opponent now attacks the other way, so the danger to you is that spot read from his side of the pitch (reflect x to $105 - x$, same y). A turnover in your own build-up is punished hard; deep in their half it costs almost nothing
- ▶ The effect: **a risky 40 m ball into a dangerous zone** has a big reward, but also a low completion chance and a heavy mirrored penalty, so its **xEPV ends up modest**. The metric values ambition only when it is likely to come off
- ▶ The two terms are **weighted 1:1 (turnover scale = 1.0)**: a turnover costs exactly what the completion would have gained, at the mirror. We kept it 1:1 on purpose, for simplicity and because **robustness checks** show the scale **moves the level** of the numbers but **not the ranking**; learning that weight is an interesting future direction

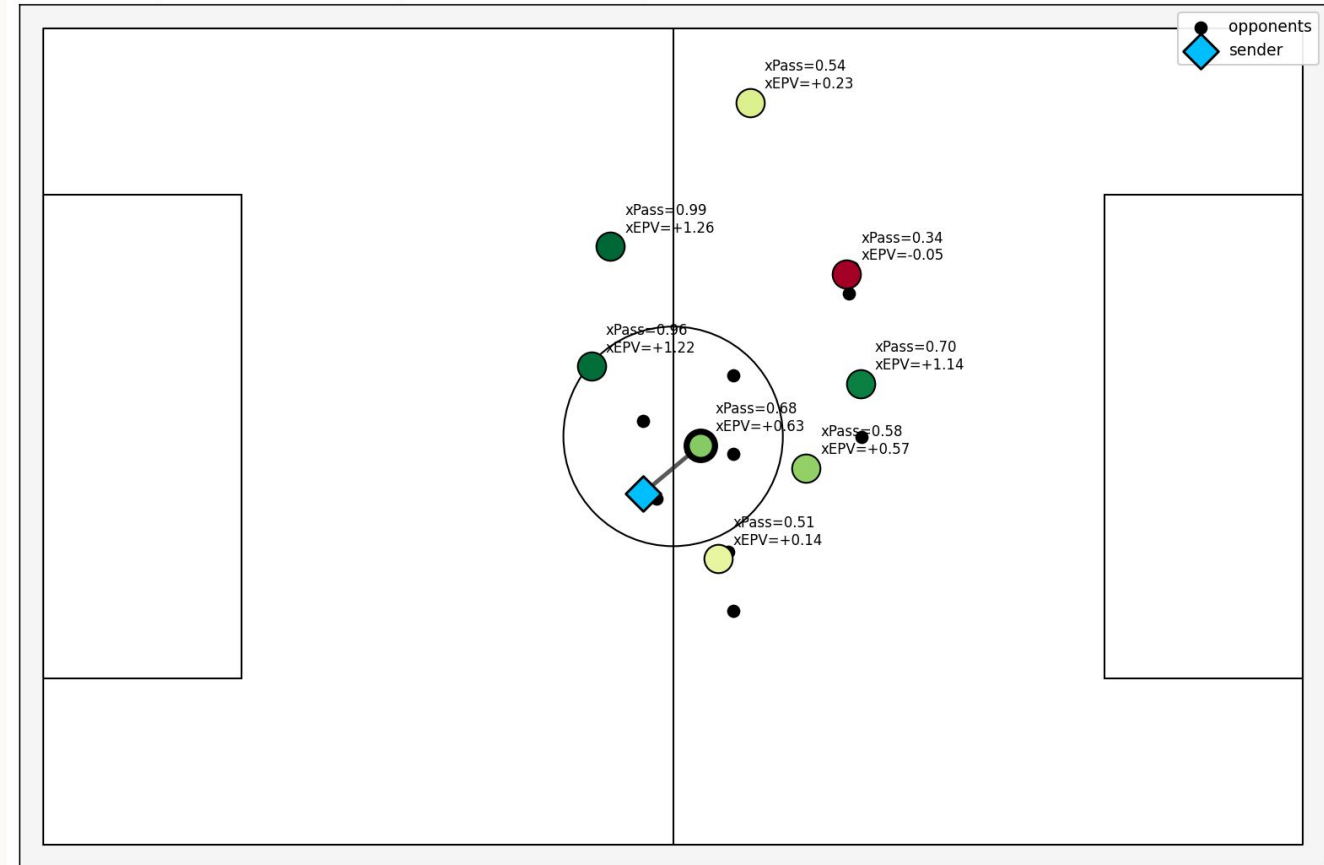
$$\text{xEPV} = \text{reward} - \text{mirrored turnover cost}$$



xEPV on the same frame: the ranking flips

- ▶ The exact **Kroos** frame from two slides ago, now **coloured by xEPV** (red negative, green positive) instead of completion probability
- ▶ The puzzle resolves: **Kroos's pick** (xPass 0.68, 4th of 8 on probability) carries **xEPV +0.63**. A medium-probability ball into an advanced zone often beats a near-certain ball into a dead one
- ▶ This is what xEPV is for: **probability times destination value**, minus the mirrored failure cost, so a safe ball into a low-value area no longer looks like the best option
- ▶ **Grading the chosen pass** against that menu of in-frame options is precisely what H2 does next

Toni Kroos (Germany) min 16' actual: completed
8 candidates | chosen xEPV = +0.63 | best alt xEPV = +1.26 | rank 4/8



ACT 3

The four studies

The heart of the work: each metric, the football it surfaces, and the validation behind it

H1 · SPACE CONTROL & VALUE

What a player does with space

Four within-role indices: how he progresses, threatens, receives, and pulls the defence

Progression — how the index is built

- ▶ We build it as a **composite**: the average of these five within-role percentiles. No single variable is progression, so we let them indicate it together
- ▶ The idea behind the three line-breaker lenses: breaking a line can mean two different things, a **bold pass** that splits bodies, or a **high-value pass** that creates danger. Rather than pick one definition, we measure both and their clean overlap
- ▶ That is why a player can reach the top two ways: as a **high-volume metronome** (*Kroos, Modrić*) or as a **low-volume vertical specialist** whose few passes are bold and valuable (*Alexander-Arnold*)

Mother variable	What it captures
LB Geom / 90	Per 90: passes that beat at least 3 opponents inside a 5 m corridor along the pass line
LB Quality / 90	Per 90: passes whose EPV added clears the player's role 75th-percentile threshold
LB EPV / 90	Per 90: passes that are both, bold geometry and high value at once
Hull Penetr. / 90	Per 90: completed passes that end inside the opponents' shape (the convex hull)
Def. Bypassed Avg	Average number of opponents a progressive pass takes out of the play

Progression — who tops it, by role

- **Progression captures** how often, and how cleanly, a player carries the ball **through opposition lines**. Read each column on its own: every score is a **percentile within that role**
- Among the centre-backs, **Bastoni leads at the 94th**: a defender whose line-breaking output reads like a midfielder's, the **modern ball-playing** profile. Among the midfielders, **Alexander-Arnold tops it (95th)** as a **vertical specialist**, just ahead of the **Kroos–Modrić metronomes**, two different ways to be progressive

H1 · PROGRESSION — top 3 per role (within-role percentile)					
CB	FB	MID	CAM	WIDE	FW
#1 A. Bastoni 94	#1 Andrija Zivkovic 90	#1 T. Alexander-Arnold 95	#1 Pedri 80	#1 Bukayo Saka 86	#1 Arda Güler 84
#2 Joachim Andersen 92	#2 João Cancelo 88	#2 Luka Modrić 94	#2 Kevin De Bruyne 80	#2 F. Conceição 85	#2 Wout Weghorst 84
#3 Fabian Schär 86	#3 O. Zinchenko 87	#3 Toni Kroos 93	#3 Dušan Tadić 74	#3 Lamine Yamal 84	#3 G. Mikautadze 81

Dangerousness — how the index is built

- ▶ Unlike the others, this is a **magnitude, not a composite**: the headline is simply the within-role percentile of total EPV added per 90
- ▶ These four are that total split by where on the pitch the value is created, and they add back up to it. **Averaging their percentiles would hide that they sum**, so we percentile the honest total and keep the four only to explain it
- ▶ EPV added is the **lift in scoring probability** from where the ball starts to where it lands: a 30 m square ball adds almost nothing, a 10 m pass that splits the block adds a lot. **The four parts show where a player's value comes from**, a centre-back through Out-Circ and Penetration, a number ten through In-Circ and Exit

Mother variable	What it captures
EPV Penetr. / 90	Per 90: value added by passes that enter the opponents' block
EPV In-Circ / 90	Per 90: value added by passes kept inside the block
EPV Exit / 90	Per 90: value added by clean passes out of the block
EPV Out-Circ / 90	Per 90: value added by passes circulated outside the block

Dangerousness — who tops it, by role

- ▶ **Dangerousness captures** the magnitude of **expected value a player creates per 90**, regardless of how many touches it takes. Each column is a **percentile within its role**
- ▶ Among the centre-backs, **Rüdiger sits at the very top** (98–100th): not a recycler but a defender who genuinely passes forward into the block. Among the forwards, **Güler** or **Conceição** stands out on **value created rather than volume**, the kind of name a raw pass count would never surface

H1 · DANGEROUSNESS — top 3 per role (within-role percentile)					
CB	FB	MID	CAM	WIDE	FW
#1 Antonio Rüdiger 100	#1 Ivan Perišić 100	#1 T. Alexander-Arnold 100	#1 Dušan Tadić 100	#1 F. Conceição 100	#1 Arda Güler 100
#2 Joachim Andersen 98	#2 Jesús Navas 98	#2 Bruno Fernandes 98	#2 Fabian Rieder 95	#2 Ousmane Dembélé 97	#2 Memphis Depay 97
#3 Maximilian Wöber 97	#3 Joshua Kimmich 96	#3 Luka Modrić 97	#3 Roland Sallai 90	#3 Mykhaylo Mudryk 95	#3 Marko Arnautovic 94

Reception — how the index is built

- ▶ We build it as a **composite**: the average of these three within-role percentiles
- ▶ The idea: it **rewards where a player stands**, not how many touches he takes. The value is in making yourself available between the lines and still keeping the ball when pressed
- ▶ **Pressure** has one definition we reuse across the whole project, **two or more opponents within 2.5 m** at the moment of release, so H1 and H2 mean the same thing by it
- ▶ This is what surfaces *Pedri* and *Wirtz*, and a cluster of ball-playing centre-backs (*Calafiori*, *Matvienko*, *Krejčíl*) that volume stats simply cannot see

Mother variable	What it captures
Between Lines %	Share of a player's touches taken inside the opponents' shape (the convex hull)
Hull Exits / 90	Per 90: clean passes played out after receiving inside the block
Press. Resist %	Pass completion when under pressure (2+ opponents within 2.5 m at release)

Reception — who tops it, by role

- ▶ **Reception captures** how often a player makes himself available **inside the opposition block**, and how well he keeps the ball when pressed. Percentiles are **within role**
- ▶ Among the attacking midfielders, **Pedri tops the pool at the 98th**, one of the standout **between-the-lines players** of the tournament. Among the wide players, **Wirtz leads** (90th): telling, because wingers usually score low here (they receive on the touchline), so the top one is the player **who drifts inside**

H1 · RECEPTION — top 3 per role (within-role percentile)					
CB	FB	MID	CAM	WIDE	FW
#1 Mykola Matvienko 93	#1 Dan Ndoye 85	#1 Robert Andrich 83	#1 Pedri 98	#1 Florian Wirtz 90	#1 Breel Embolo 89
#2 Ladislav Krejčí 91	#2 Bendegúz Bolla 83	#2 Jorginho 82	#2 C. Eriksen 80	#2 Jamal Musiala 85	#2 Ché Adams 81
#3 Nacho Fernández 91	#3 O. Zinchenko 82	#3 Granit Xhaka 80	#3 Jonas Wind 78	#3 L. Pellegrini 84	#3 Kacper Urbański 76

Gravity — how the index is built (experimental)

- ▶ We label this index **experimental**: it tries to measure something new, **how the defence reacts to a player**, rather than what he does with the ball. We show it as a **direction worth exploring**, not a finished metric
- ▶ It uses a **leave-one-out baseline**. For each zone of the pitch we work out the normal defensive shape from every other player's actions there, then measure only **how much the defence moves beyond that when this player has the ball**. So it is not just defenders are near him, it is more than the zone alone would explain
- ▶ The **three signals** look at different things: two at **how tightly the defence packs** (his 4 nearest opponents; the block's area), one at which direction its **centre of mass shifts**, in metres
- ▶ Because they are **three different behaviours**, a player can score high on one and low on another. We keep them apart on purpose, and as the next slide shows, they **often pull in different directions**

Mother variable	What it captures
Space Attraction %	How much closer the 4 nearest opponents sit than the zone baseline expects
Gravity Hull %	How much the opponents' block area shrinks vs the same baseline
Def. Pull (m)	Distance, in metres , the opponents' centre of mass slides toward the player

Gravity — who tops it, by role (experimental)

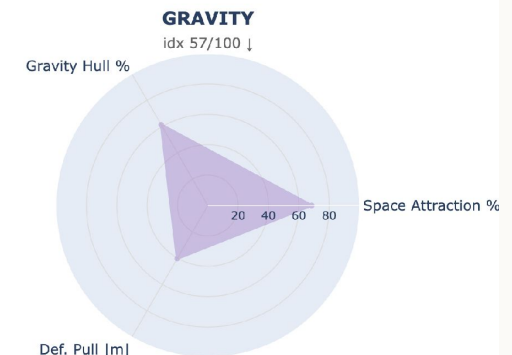
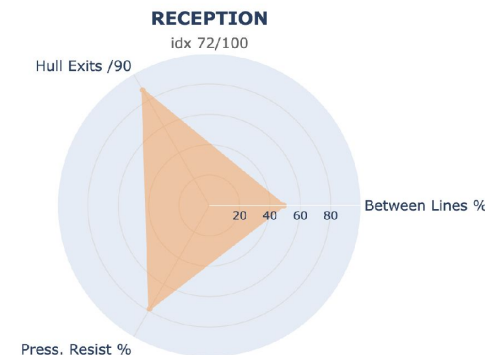
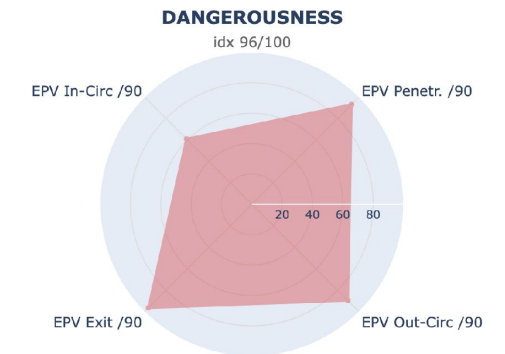
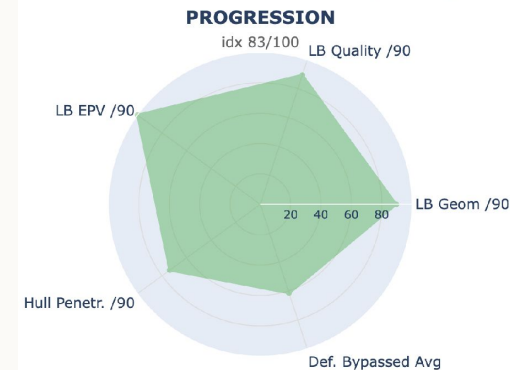
- These are the players the **experimental Gravity index** ranks highest in each role, on one tournament: *Strelec* among the forwards, *Mlakar* wide, *Baumgartner* among the tens, *Frankowski* at full-back. Read it as a **signal to look closer, not a verdict**
- Notice the top scores here (**mostly low-to-mid 90s and below**) sit **lower than** on the **other indices**, where leaders reach 98-100. That is the experimental part showing: the **three sub-signals** that make up Gravity **often disagree**, so they rarely line up to push one player to the very top

H1 · GRAVITY — top 3 per role (within-role percentile)					
CB	FB	MID	CAM	WIDE	FW
#1 Jan Vertonghen 76	#1 P. Frankowski 81	#1 G. Kochorashvili 93	#1 C. Baumgartner 78	#1 Jan Mlakar 95	#1 David Strelec 92
#2 Robin Hranac 75	#2 Otar Kakabadze 77	#2 Lukas Provod 92	#2 Dušan Vlahović 73	#2 Kenan Yıldız 84	#2 Mateo Retegui 80
#3 Ilya Zabarnyi 72	#3 Silvan Widmer 77	#3 Salih Özcan 92	#3 Roland Sallai 72	#3 Jasir Asani 82	#3 Benjamin Sesko 80

The four-index card, worked: Kimmich

- **One player, four panels**, each one a **percentile within the full-back pool**, broken down into its underlying variables
- **Kimmich: Progression 83, Dangerousness 96, Reception 72, Gravity 57**. The **shape reads like the player**: very strong at moving the ball and creating threat from full-back, solid at receiving between the lines, and more modest on Gravity (the defence reacts to where he positions himself rather than getting dragged toward him)
- This is the case for **four numbers instead of one**: a single overall score would blend these into one figure and hide the shape. Four axes keep his strengths and his quieter areas both visible, which is what a scout actually wants to see
- The four axes are not saying the same thing (they **correlate at most |0.56|**), so each panel adds information. One nice find along the way: **receiving between the lines and drawing defenders close** turn out to be the same profile ($\rho \sim 0.89$)

Joshua Kimmich · Germany · Right Back (FB) · 486 min



Honest by design: internal consistency

- ▶ *Cronbach's* alpha asks "do a composite's variables move together?"
- ▶ **Gravity at -0.03 is the honest signature of its construct:** a forward dragging a CB wide and a winger pinning both full-backs are both 'gravitational' but doing different things, so their variables don't co-move

Index	Cronbach's alpha	What it means
Progression	0.77	Tight construct: the 5 variables genuinely measure one dimension
Reception	0.41	Moderate composite (high on CB/CAM, lower on FW small samples)
Dangerousness	0.32	Expected low: 4 additive EPV components, not a reflective composite
Gravity	-0.03	Negative by design: 3 different pulling behaviours , not one skill

Contextual vs naive: the re-ordering proof

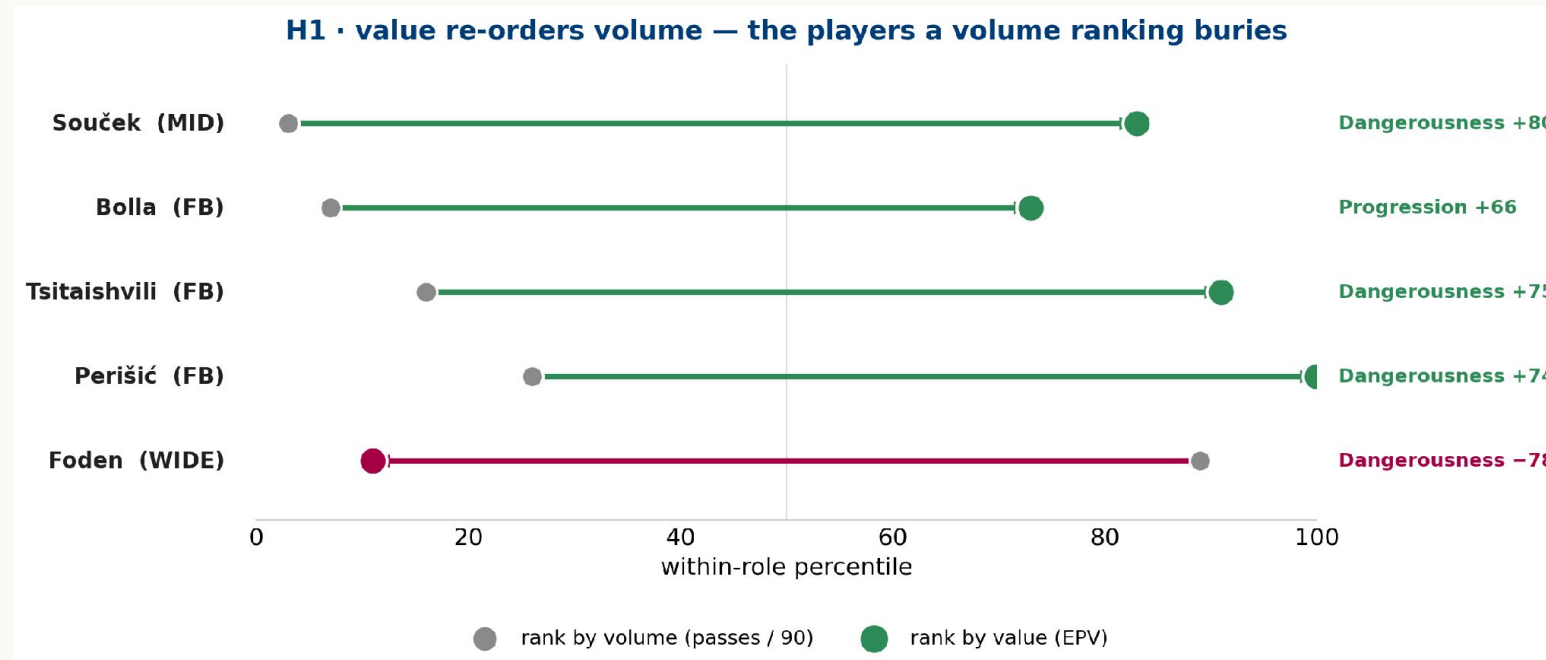
► **The test of H1's core claim:** rank all 272 players by a **naive volume proxy**, then by our **contextual index**, and measure how far the ranking moves (within role). **Low correlation + a large share of players shifting = volume was a poor proxy for value**

Index	Naive proxy	Spearman ρ	Mean $ \Delta $ pctile	% shifting > 20
Progression	passes / 90	0.47	21.7	47%
Dangerousness	passes / 90	0.28	27.8	54%
Reception	between-lines %	0.75	14.9	29%
Gravity	proximity %	0.60	18.5	39%

- Read ρ as agreement. **Dangerousness at 0.28:** volume and value barely agree on who is dangerous, and **54% of players move 20+ percentile** points. The ranking is genuinely re-ordered, and the **new order is football-coherent**, not just noisier
- Progression and Dangerousness are tested against **plain passing volume**; Reception and Gravity against their **own most intuitive raw variable**. Either way the contextual index moves the ranking

Value re-orders volume: the discoveries

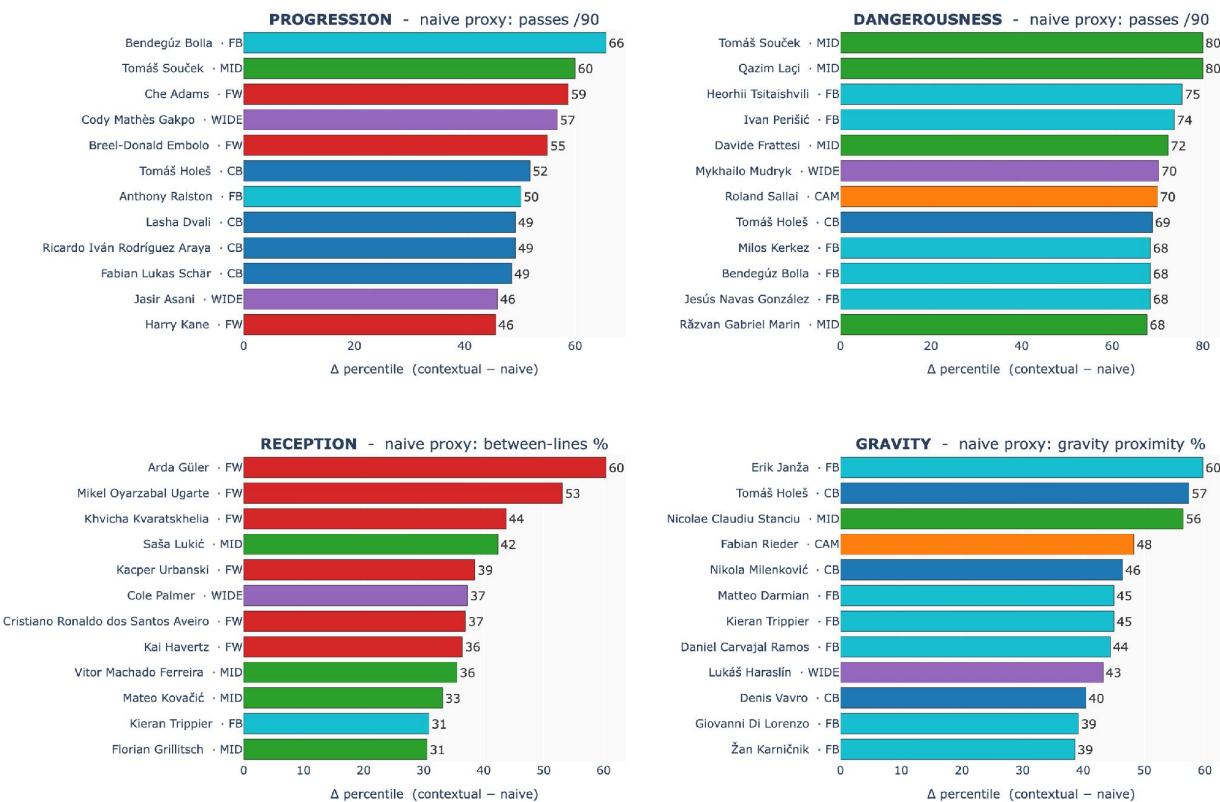
- ▶ **Souček**: 3rd percentile by passes/90, 83rd by EPV created (+80). Almost no volume, but every touch moves the needle
- ▶ **Bolla**: 7th by volume, 73rd on Progression (+66). The few passes he plays are clean line-breakers
- ▶ **Foden**: 89th by volume but 11th by EPV (−78). Heavily involved, yet he adds little value in this sample
- ▶ Two players the volume ranking buries, one it inflates



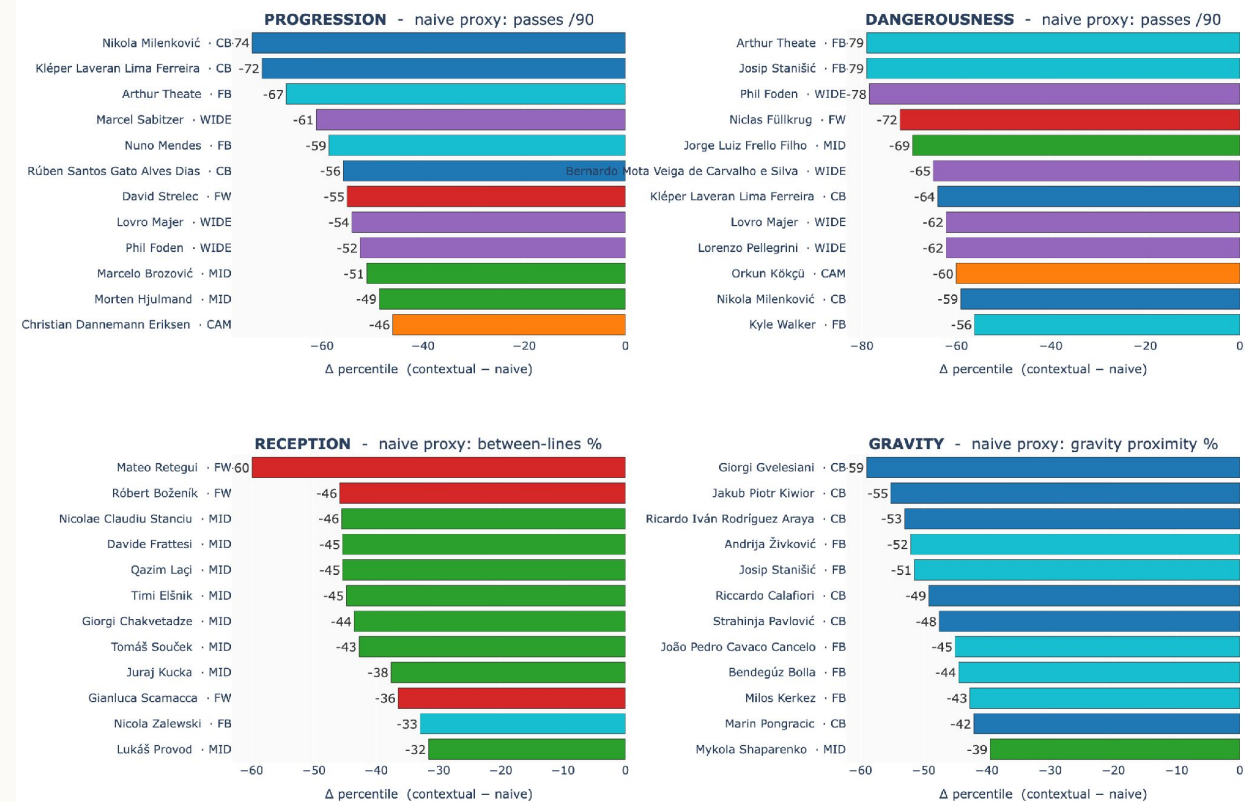
The full picture: who volume misses, and who it inflates

- The same Δ test across all four indices, both directions (Δ = contextual – naive percentile, within role, coloured by role). Left: the 12 biggest positive moves per index, players volume buries (*Bolla* +66, *Souček* +80, *Güler* +60). Right: the 12 biggest negative moves, players volume inflates (*Foden* –78 on Dangerousness)

Scouting discoveries - players surfaced by contextual indices but missed by naive stats



Naive overrating - players whose naive rank is inflated by team context



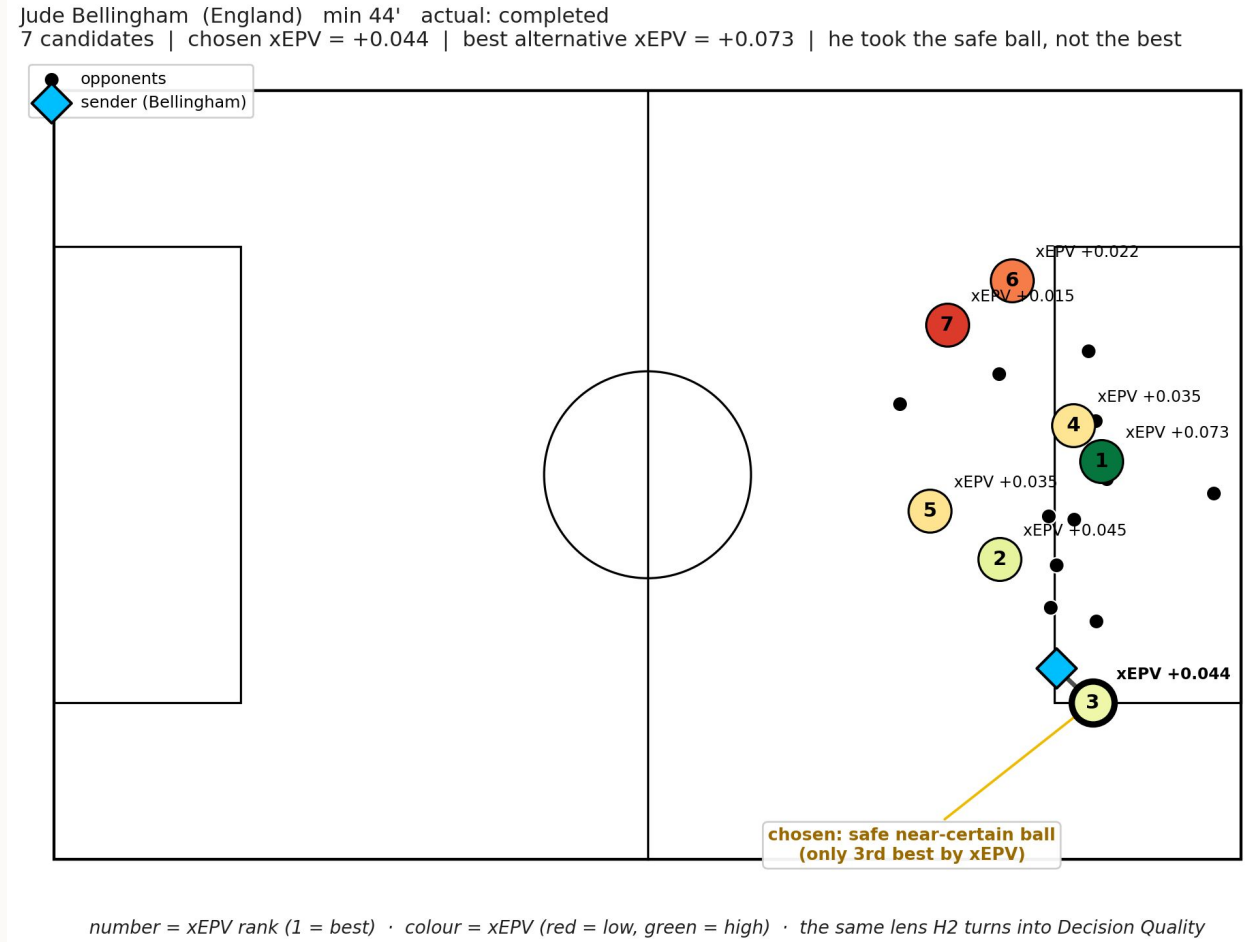
H2 · DECISION QUALITY

Was it the right pass?

Not the value of the pass played, but its value against the in-frame options ignored

Scoring the decision, not the event

- ▶ A pass can add value and still be the wrong choice, if a better option was open and ignored. **Standard models score the event; we score the decision**
- ▶ The **DQ Score**, in words: **of all the in-frame options, what share had an xEPV at or below the pass he actually chose?** Average that over his decisions, then take the within-role percentile
- ▶ **Why a within-event rank**, and not the obvious alternatives: **not absolute xEPV** (that would reward whoever plays in high-value zones, by position); **not regret-vs-best** (tested and rejected, it punishes players who operate in option-rich frames)
- ▶ It asks the only fair question: **given what was actually on offer in this frame, how good was the pick?** The pitch-zone value cancels inside each event
- ▶ **The frame**, in numbers: his **chosen ball**, $xEPV = 0.99 \cdot 0.044 - 0.01 \cdot 0.007 = +0.044$; the **best option**, $xEPV = 0.56 \cdot 0.136 - 0.44 \cdot 0.008 = +0.073$. He took the **near-certain ball over the more valuable one**, exactly the spread DQ measures

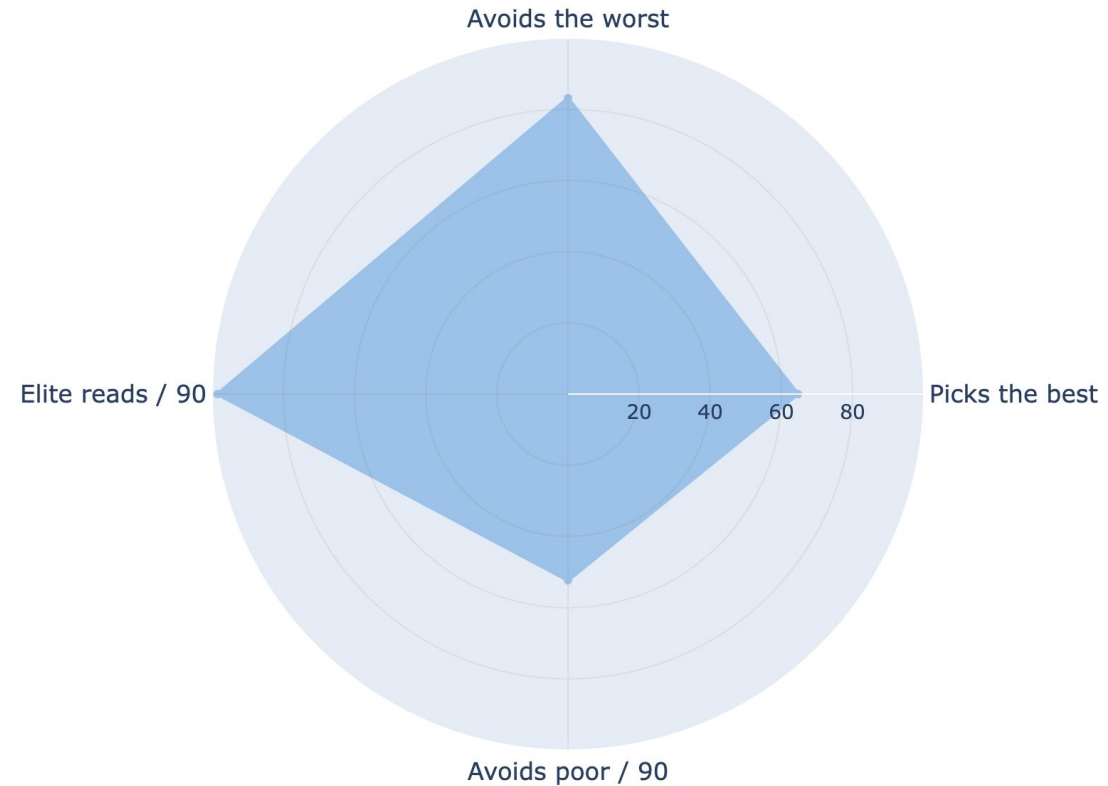


Rebuilding the menu the player faced

- ▶ For every pass, we take each teammate who was in the frame and score the pass he could have received, with the same xPass then xEPV. That **rebuilds the set of options** the player actually had: **222,274** in **total**, about **7 per pass**
- ▶ **We leave** out three kinds of option, each for a clear reason: **headers** (a different kind of pass), **goalkeepers** (different job), and **balls into space with no teammate near the landing spot** (no clear 'who was he passing to?')
- ▶ For each option we also have to **guess its height**. Simple rule: **under 30 m we treat it as a ground pass, over 30 m as a lofted one**, because that is how real passes behave (almost all short ones are on the ground, almost all long ones in the air)
- ▶ There is **genuinely something to grade**: players pick the single best available option only **15% of the time**, about one in seven. If everyone always picked the best, the index would be flat and useless
- ▶ The radar (right) breaks the **single DQ score into four**: how good the picks are (**Picks-best, Avoids-worst**) and how often they come (**Elite reads/90, Avoids-poor/90**)

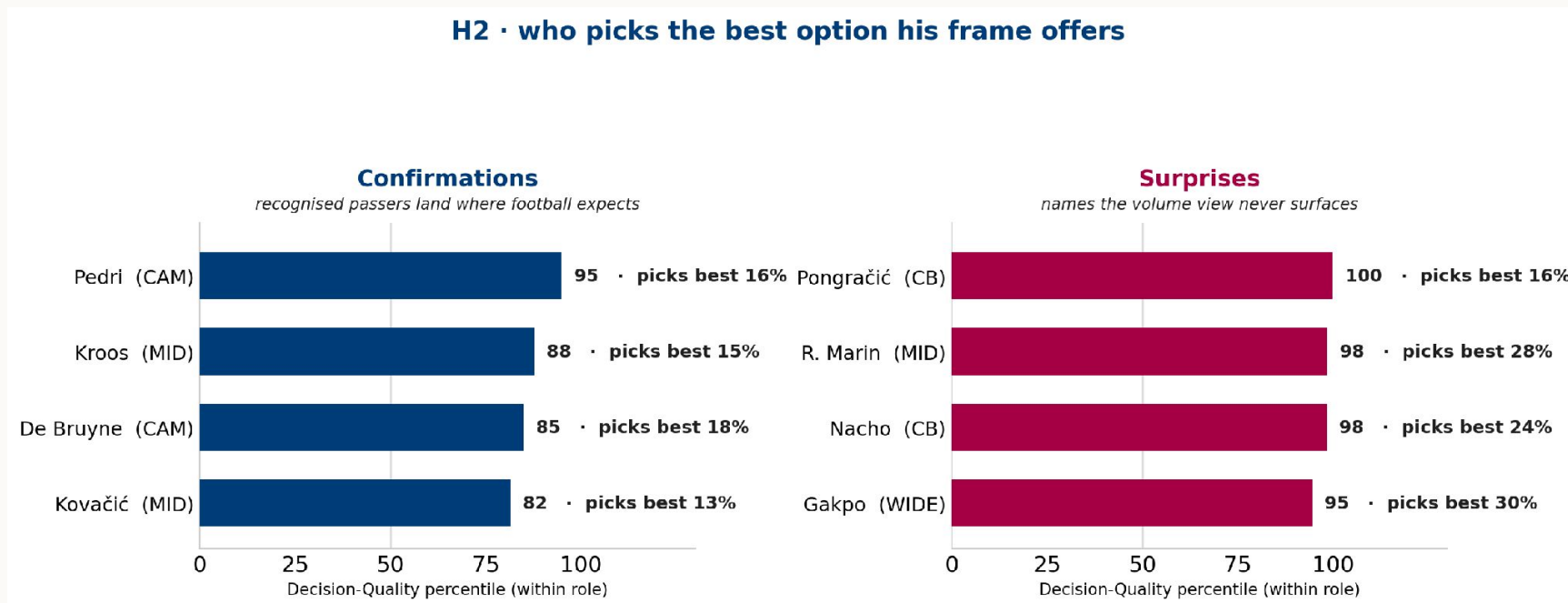
DECISION QUALITY · idx 88/100

Toni Kroos · Germany · Left Defensive Midfield (MID) · 485 min · n = 379 decisions



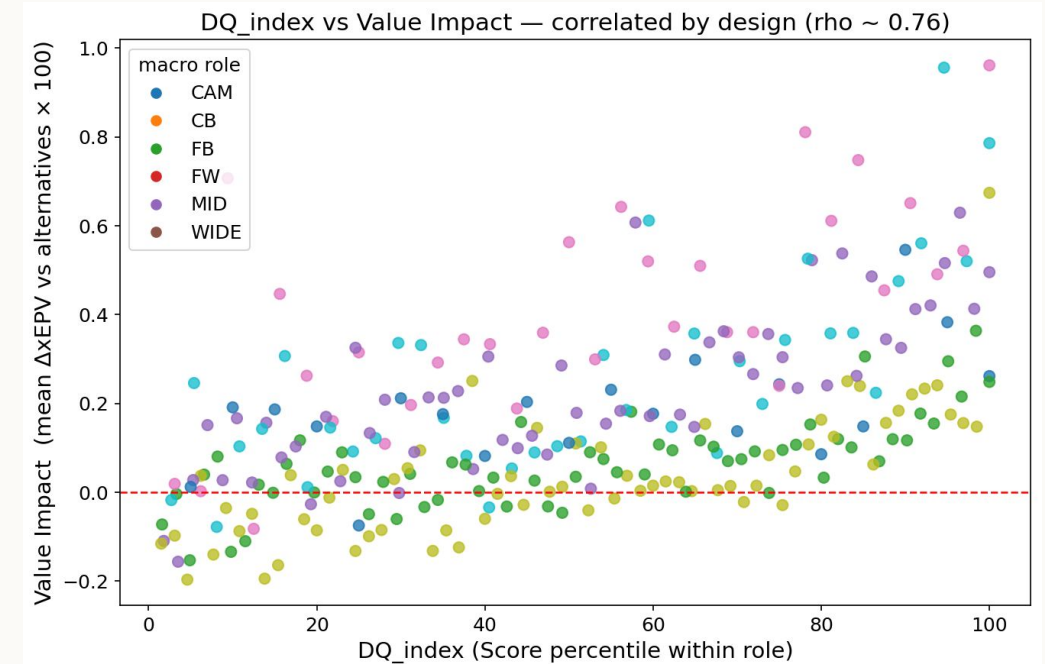
Who actually picks the best option

- **Confirmations:** *Pedri* (95th), *Kroos* (88th), *De Bruyne* (85th). The **recognised passers** land where football expects
- **Surprises:** *Nacho* (98th), *R. Marin* (98th), *Gakpo* (95th). **Names a volume view never surfaces**
- The variable '**picks best %**' is how often the chosen lane was the **top option in the frame**
- The **whole Spain scores high**, one of the **confirmation of the tournament**



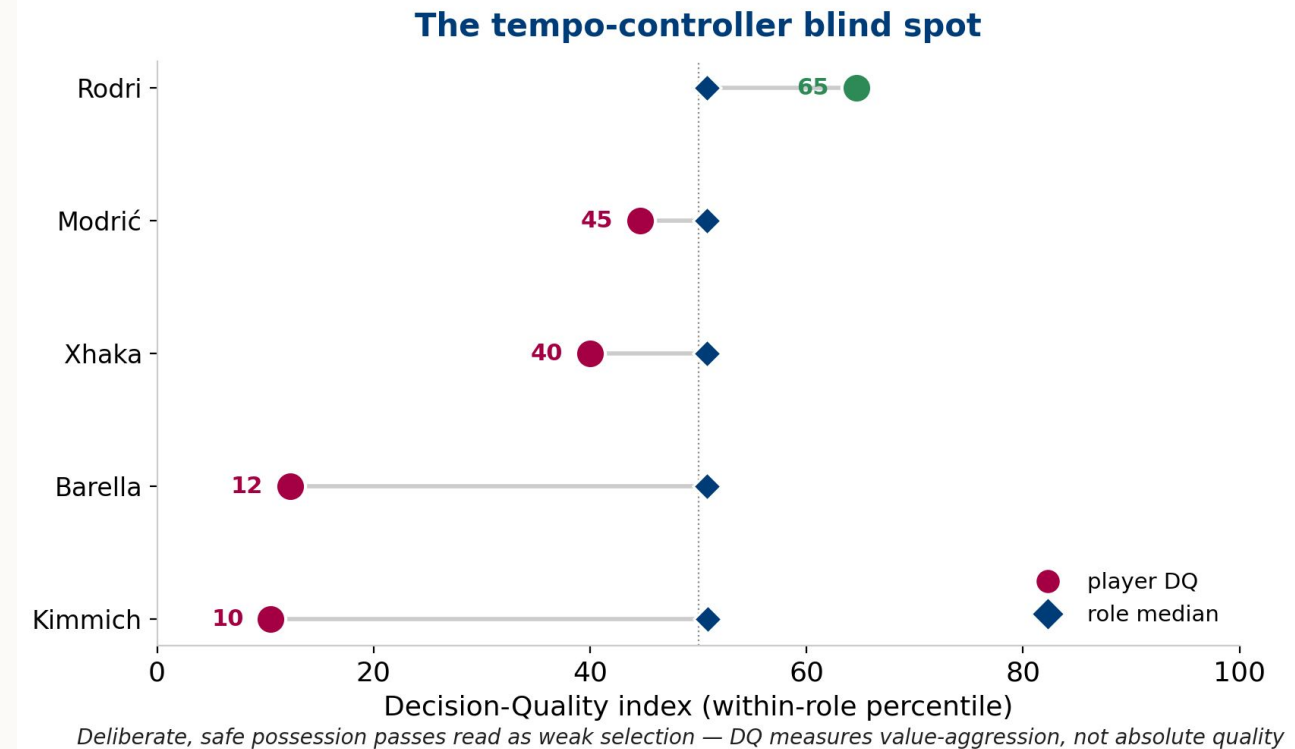
Does the ranking hold? Three checks

- **Robustness.** We refit the whole pipeline, **moving one knob at a time**: **turnover scale** (0.5/1.0/1.5), **minutes floor** (90/135/180), **minimum alternatives** ($\geq 1/\geq 2/\geq 3$). The player order holds at **Spearman rho ≥ 0.95 every time**, so the knobs set the level of the scores, not who ranks above whom
- **Spatial sanity.** Mean chosen **xEPV rises smoothly with the sender's pitch position**, exactly as it should. Split by completion probability, the riskiest passes carry both a big reward and a big mirrored penalty; the safe ones are almost pure reward
- **A companion metric agrees (figure).** **Value Impact**, the average value a player's pass adds over his alternatives, tracks the decision-quality percentile at **rho ~ 0.76 within role**: the **same signal, value-weighted**



Honesty: the tempo-controller blind spot

- ▶ *Kimmich, Modrić, Xhaka and Barella* sit **below their role median on DQ**
- ▶ A deliberately safe, **possession-securing pass** reads as **weak selection**. The model has **no game state** (no scoreline, no clock, no instruction)
- ▶ So **DQ** measures the **value-aggression of pass selection within the reachable action set**, not absolute decision quality
- ▶ The gap is in the lens, not necessarily in the player



H3 · OFF-BALL MOVEMENT

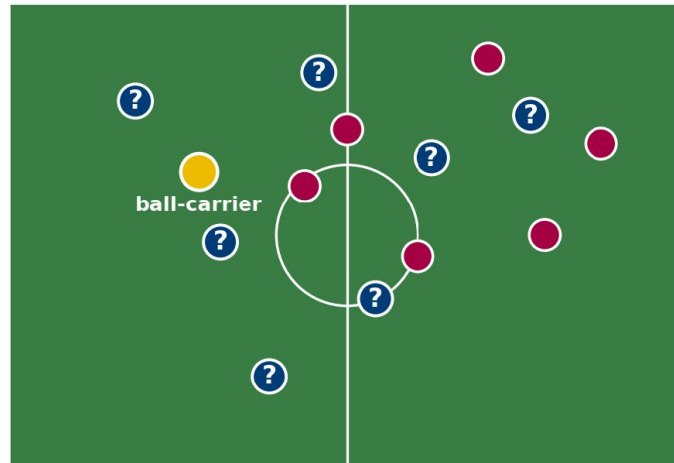
The run that was never served

The dangerous off-ball space a player offers that his teammates fail to use

The identity problem

- ▶ A 360 frame shows **where players are, not who they are**
- ▶ Only the **ball-carrier** is **named**. Everyone else is an anonymous **team-coloured dot**
- ▶ This is why **off-ball** rating is normally considered **impossible on public data**

What a 360 frame gives you



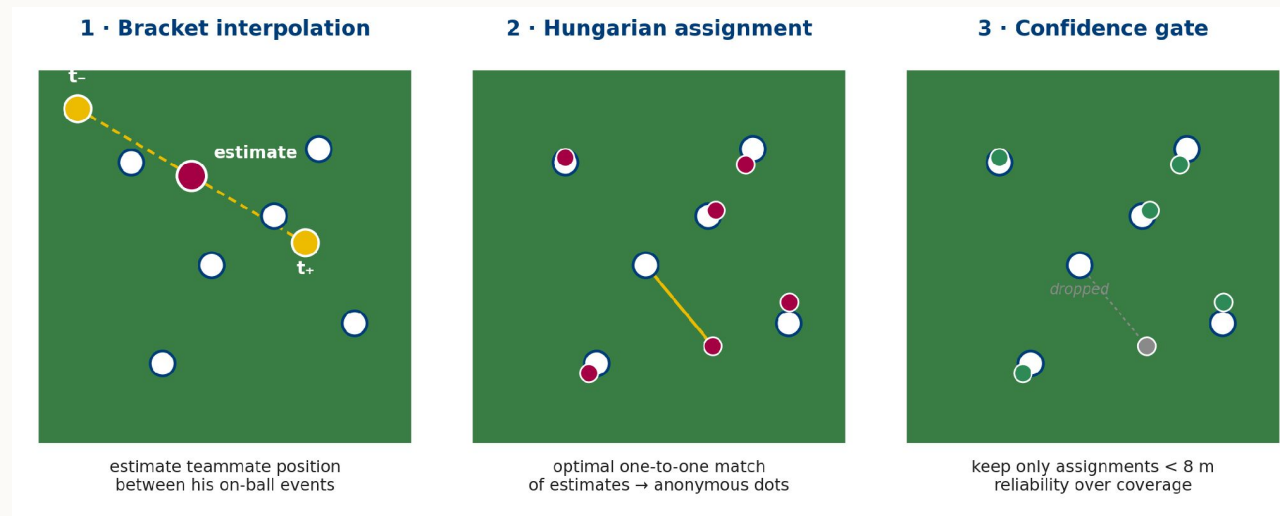
After the receiver resolver



Only the ball-carrier is named — everyone else is an anonymous team-coloured dot

The receiver resolver: how we put names on dots

- ▶ A **360 frame** names only the ball-carrier. To credit off-ball work we first have to infer who each anonymous dot is. Three steps:
 - **Estimate position by bracket interpolation.** For each teammate we find his nearest named on-ball event before and after the frame, and read off where he must have been at the pass instant. This **beats the naive last-known position**, which goes stale the moment he keeps running
 - **Match estimates to dots with the Hungarian algorithm:** the one-to-one pairing that **minimises the total distance** over all pairs at once. A greedy nearest-dot rule can trap a later estimate onto a far wrong dot
 - **Keep only confident assignments:** within 8m, discard the rest. **Reliability** is more important than coverage in this case

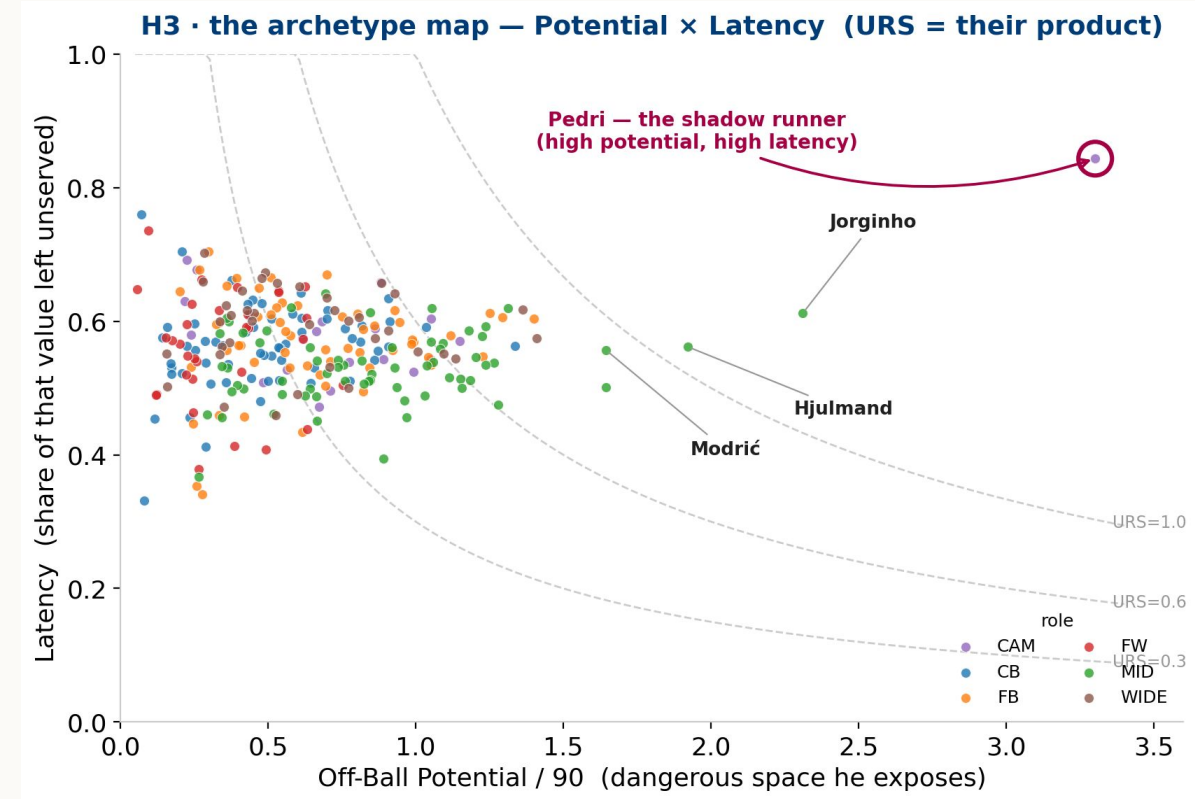


Does it work? We tested it against the truth

- ▶ We tested it. Completed passes name their real recipient, so **we hid that name**, asked the **resolver to pick the dot**, and checked. It is **right 93% of the time** on the assignments it keeps
- ▶ **Calibrated by parameter sweeps**: A **5s staleness window** and **8m confidence radius** were selected via a 10-match sweep, finding the exact sweet spot where accuracy reaches its peak without losing data coverage
- ▶ We accept a proxy assumption. We cannot directly validate off-ball runs due to a lack of ground truth. However, our **93% accuracy on passes justifies the jump to individual-level rankings**; a lower score would have forced a fallback to safer, aggregate zone numbers
- ▶ We are honest about **one bias**. **Midfielders are simply seen more often** (more touches near them, easier to place), so they fill the top of the raw board. That is about who gets recognised, not who is best, which is why **every ranking here is read within role**

URS: the run that was there but never served

- ▶ While a teammate has the ball, every other player in the frame is a run he could be served. For each we ask two simple things: **how much would that pass be worth** (its xEPV), and **was he actually played in** (did the ball reach him within 4 seconds)?
- ▶ **URS** adds up **the value of the runs that were not served**, per 90 minutes. We use **4 seconds** because that is how long a real pass takes to arrive (almost all land within 3), so it **counts genuine misses** without catching the next phase of play
- ▶ It splits in two: **URS = how much value he offers (Potential) × how much of it goes unused (Latency)**. A shadow runner offers a lot and gets served little, the top-right of the chart
- ▶ We call it uncapitalised because served runs already show up in the data. **What the event log misses is the dangerous run a teammate offered and nobody used**



Honest by design: what URS is, and isn't

► What it is :

- **Potential** and **Latency** are **nearly independent** within role ($\rho \sim 0.08$): how much dangerous space a player exposes is almost unrelated to whether his team uses it. That near-zero number is the **empirical heart of the shadow-runner idea**
- And **mostly a player signal**: a player's **Latency vs his teammates' mean H2 decision quality** correlates at only $\rho \sim -0.13$ (right direction, but small correlation)

► What it isn't :

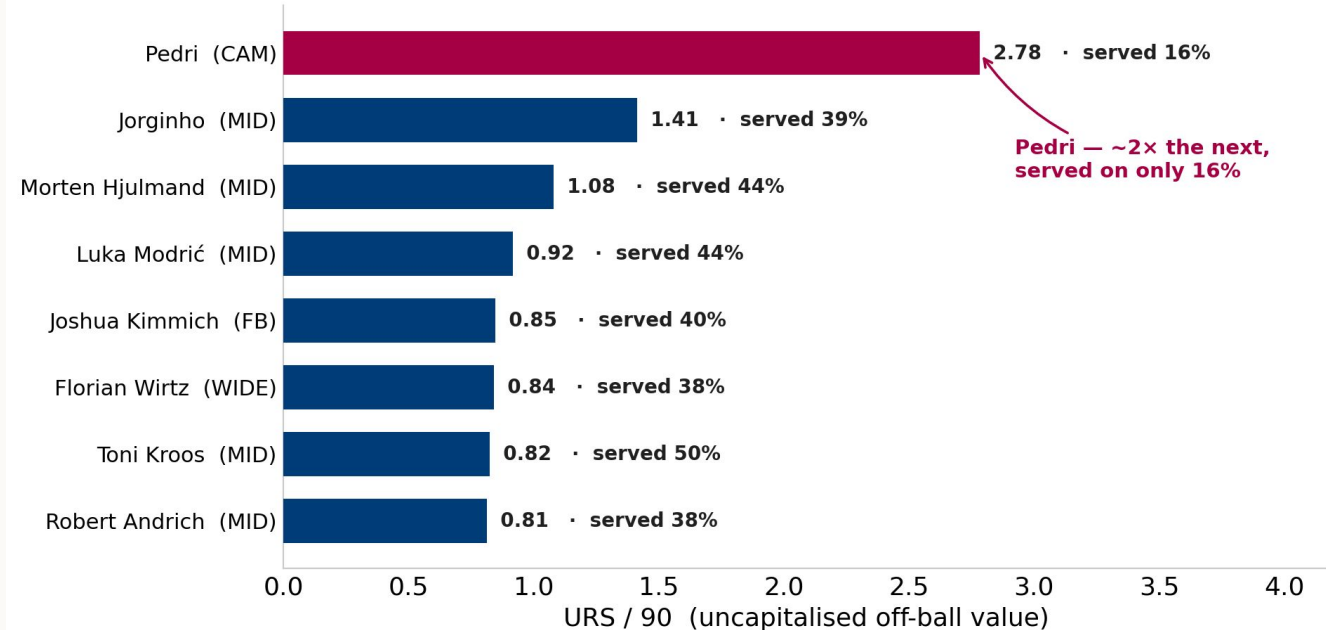
- **Not a fully independent radar**: **Off-Ball Potential** tracks the URS headline at $\rho \sim 0.97$ by construction ($URS = Potential \times Latency$). We keep it as the readable volume axis
- **Not purely individual**: it is **conditional on teammate execution**. So read Potential for the cleanest individual signal, and the headline as off-ball value in his current team context

► **The hook**: this is why **URS complements H2**. H2 grades the passer; Latency captures a player's off-ball profile. A **high-Potential, high-Latency player in a rigid or low-expression system = value systematically wasted** (the actionable scouting signal to find hidden gems for high-creativity teams)

The shadow runners of Euro 2024

- ▶ **Pedri leads at URS/90 of 2.78**, about **twice the next man**. Then *Jorginho*, *Hjulmand* and *Modrić*: playmakers offering runs their teams under-use
- ▶ The variable '**served %**' is the **share of offered value actually played**. *Kroos* uses 50%, *Pedri* only 16%: the trusted threat vs the shadow runner, same exposure, opposite outcome
- ▶ **Read within role** (the algorithm naturally recognises midfielders more often due to central event density), the right archetype tops every pool:
 - CB: *Tah*, *Rüdiger*, *Calafiori*
 - FB: *Kimmich*, *Dalot*, *Trippier*
 - CAM: *Pedri*, *Gündoğan*, *Eriksen*
 - WIDE: *Wirtz*, *Conceição*, *Musiala*

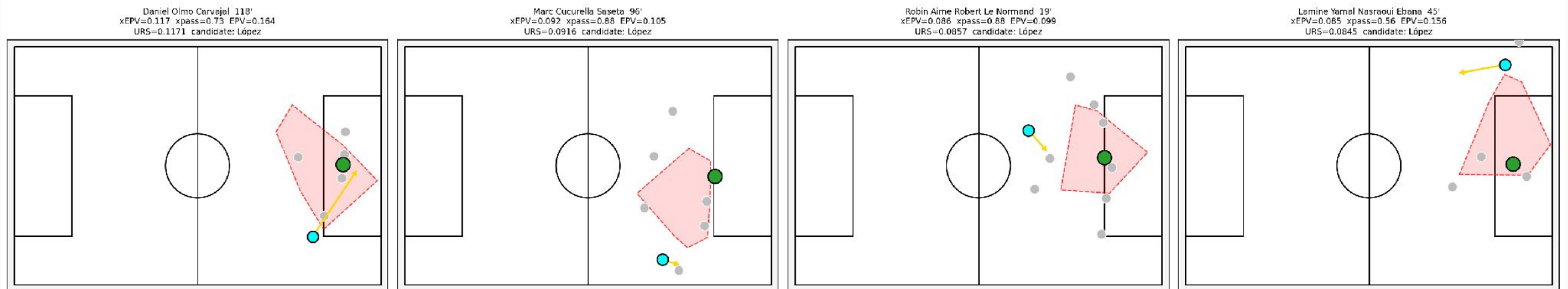
H3 · the shadow runners — most off-ball value left unserved



Pedri: the clearest case of the tournament

- ▶ **URS/90 of 2.78**, roughly twice the next player in the pool
- ▶ He offers **more off-ball value than anyone** (Potential 3.30/90), and is **served on only 16%** of it
- ▶ Honest flag: **185 minutes**, about two matches, so **an extreme signal on a thin sample**
- ▶ It holds up at pool level: **split-half correlation of 0.72** across the tournament

Top-4 uncaptialised runs — Pedro González López



H4 · PLAYER SIMILARITY

Who plays like him, for less?

Match players by their contextual style DNA, then surface cheaper look-alikes within role

Player similarity: the spatial DNA

► Real footballers are never identical nor total opposites, so genuine pairs sit in a ~23-88% band. We re-rank every axis to a within-role percentile first, else H1's tighter-spread indices would under-weight (Gravity at 3.7% instead of 1/11), and the single closest match would flip for 26% of players

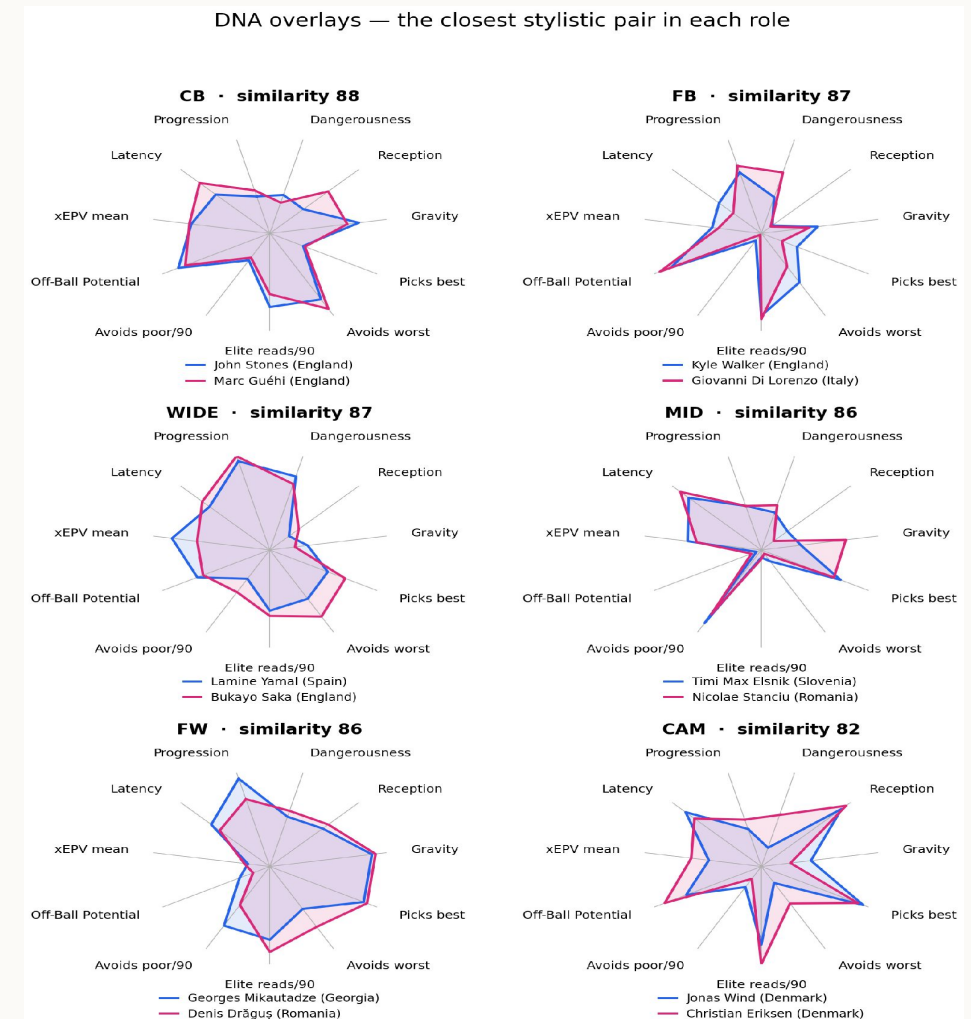
► Each player is an **11-axis DNA** assembled from the earlier studies: the **4 H1 spatial indices** + the **4 H2 decision axes** + the **3 H3 off-ball axes**. A style fingerprint: **how he plays**, not his team or his position label

► We deliberately **exclude the two headline aggregates** (H2's DQ index, H3's URS/90): each is just a function of axes already in the vector, so including them would double-count the same signal

► **Distance is Euclidean**, then mapped to a 0-100 similarity:

$$\text{Similarity} = \left(1 - \frac{d}{D}\right) \times 100, \quad \text{with } D = \sqrt{11} \times 100 \sim 331.7$$

The largest distance two percentile vectors could ever have. Fixing D to that theoretical max (not the closest/furthest observed pair) makes the score absolute: it depends only on the two players, so adding or removing others never shifts it



The closest stylistic pairs of Euro 2024

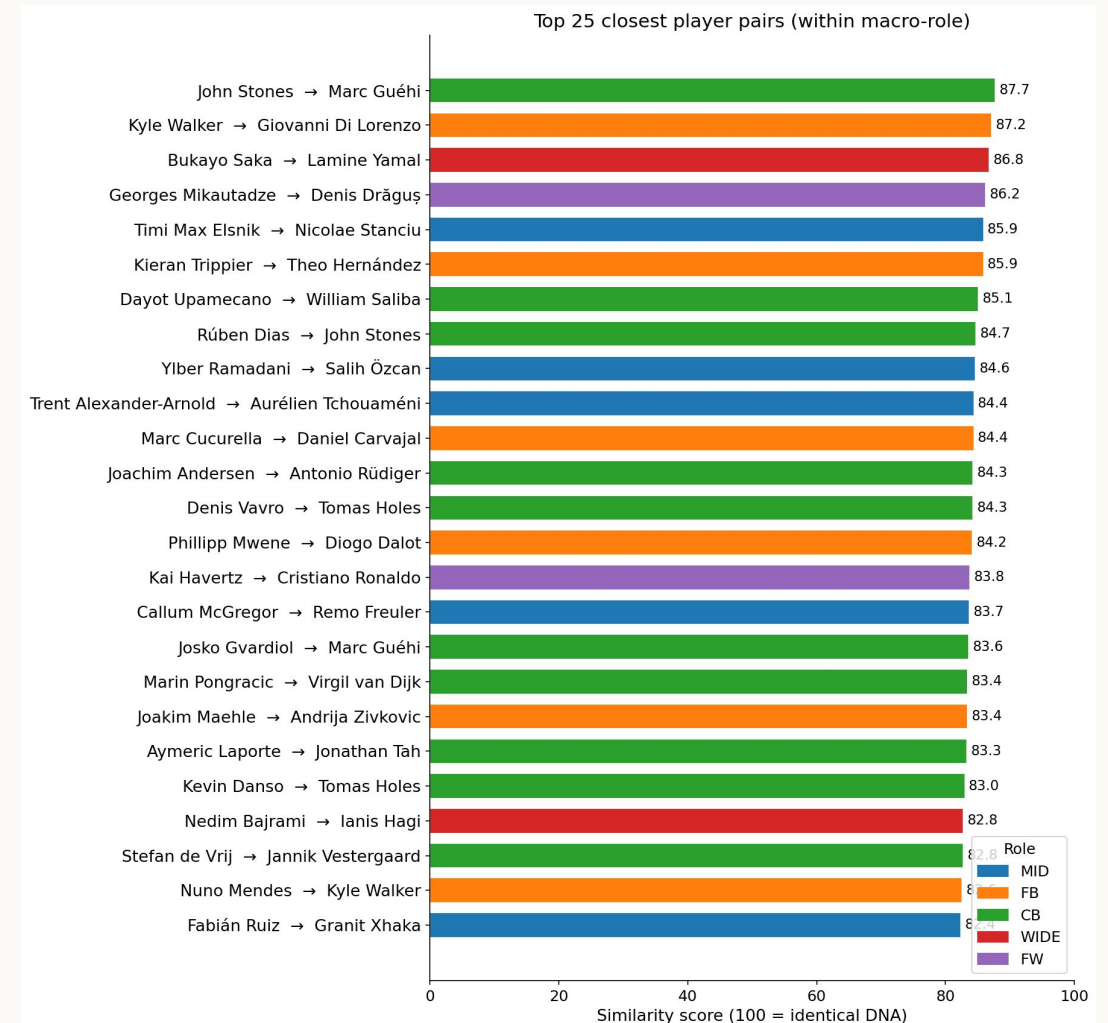
► **Coloured by macro-role:** The engine **pairs players who play alike**, not who line up alike. For example: *Trent Alexander-Arnold* played as a central midfielder and *Aurélien Tchouaméni* as defensive midfielder, but they still share a spatial DNA score of 84.4

► The **tightest DNA matches**, within macro-role:

- *Stones* and *Guéhi* (87.7), *Walker* and *Di Lorenzo* (87.2)
- *Saka* and *Yamal* (86.8), *Cucurella* and *Carvajal* (84)
- *Alexander-Arnold* and *Tchouaméni* (84.4), *Fabián Ruiz* and *Xhaka* (82.4)

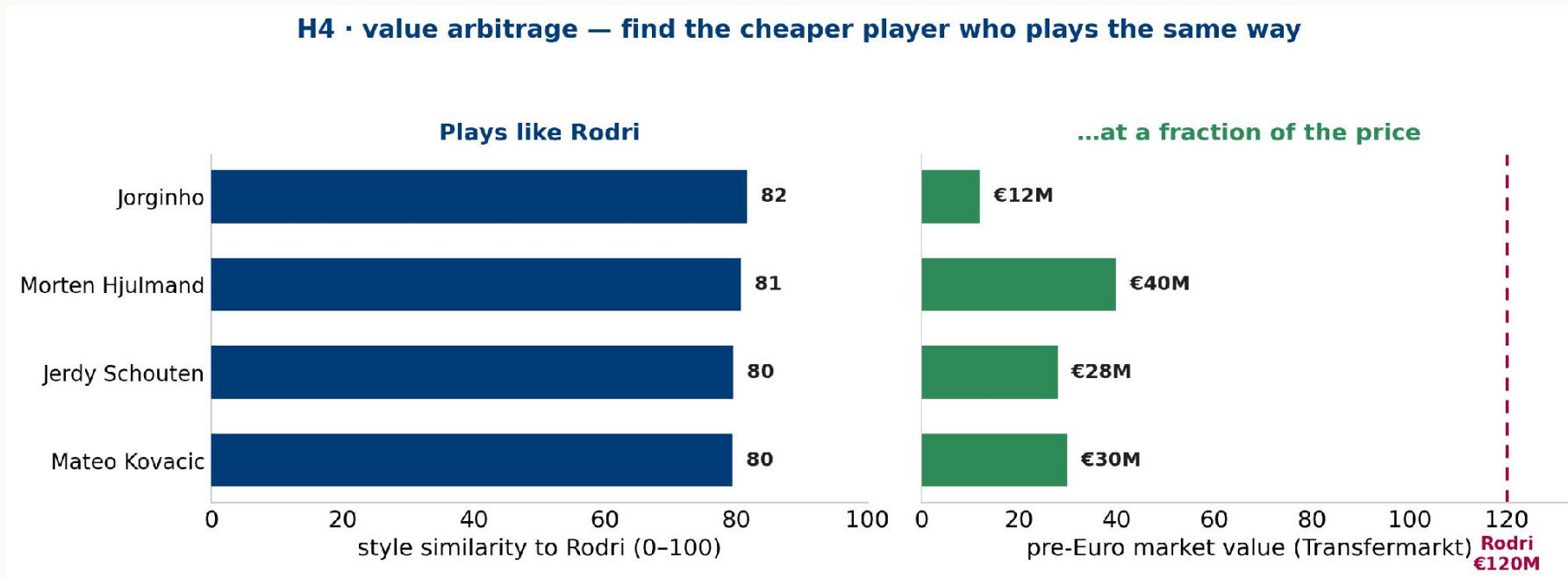
► Specific **case studies** for in-depth analysis (rounding the numbers):

- *Rodri* shares the most spatial DNA with these players:
Jorginho (82) - *Hjulmand* (81) - *Schouten* (80) - *Kovačić* (80)
- *Gündoğan* shares the most spatial DNA with these players:
Xavi Simons (76) - *Bellingham* (76) - *Kökçü* (74) - *Dani Olmo* (72)



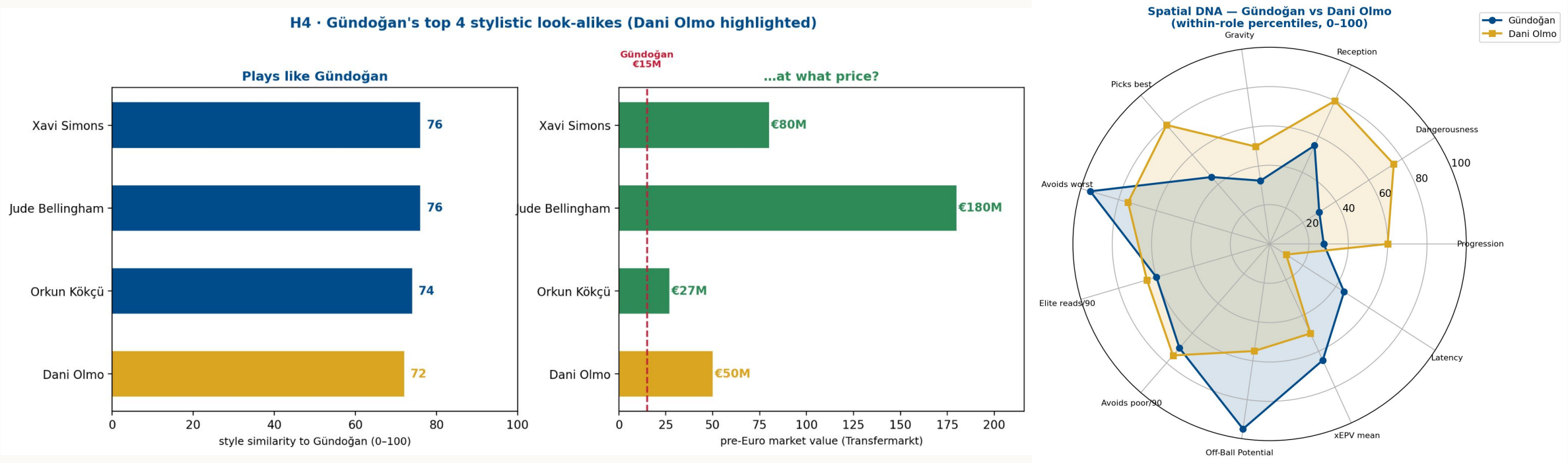
Value arbitrage: the hypothetical cheaper look-alike

- ▶ For a top-tier target, surface his cheaper stylistic look-alikes, by style and not by position label
- ▶ *Rodri* (pre-Euro €120M) plays like *Jorginho* (similarity 82, €12M), *Hjulmand* (81, €40M), *Schouten* (80, €28M) and *Kovačić* (80, €30M): the deep-lying alternatives a club could sign for a fraction of the price
- ▶ The dashboard shows both prices plus age: the pre-Euro value is the true arbitrage figure (what you could have paid before the tournament revealed him); the post-Euro value shows where the market moved, so a scout sees if a cheap look-alike has already started to appreciate



Real case scenario: *Gündoğan* replacement

- ▶ Context: Barcelona’s midfielder *İlkay Gündoğan* left the club right after EURO2024, *Dani Olmo* was the designated replacement
- ▶ *Dani Olmo* ranks 4th (similarity 72), *Xavi Simons* (76), *Bellingham* (76) and *Kökçü* (74) are actually more similar
- ▶ His value incremented post-tournament (€50M → €60M)
- ▶ One of the best alternatives among the best possible candidates with the same similarity score as *Gündoğan*



Model Validation: Statistical Rigor & Tactical Reality

- ▶ **Style, not team:** only 10.3% of nearest matches are teammates. If the metric were quietly reading 'same side', that share would be high
- ▶ **No single axis carries it:** dropping each of the 11 axes in turn, the top-5 matches survive 83.5% of the time. A leaner 6-axis version drops to 72.8% and pairs players who reach the same level in very different ways, which is why the full 11-axis DNA is kept
- ▶ **Axes aren't duplicates:** the within-role |Spearman| matrix peaks at 0.66 (Avoids-poor/90 vs Off-Ball Potential) and is football-sensible, with most pairs far lower. None high enough to be redundant
- ▶ **Discriminating power & Face Validity:** best-match scores spread properly (median 79, range 65-88). 14 players have a strong stylistic twin (>85), 47 are more one-of-a-kind (<75)
 - **On-pitch proof:** The engine maps intuitive tactical families, cleanly pairing *Stones* ↔ *Guéhi* (87.7), *Walker* ↔ *Di Lorenzo* (87.2), and *Yamal* ↔ *Saka* (86.8)
- ▶ **Methodological Pivot** (Why not clustering?): The original plan was K-Means archetypes. Cross-role clustering merely rediscovers macro-roles (defenders with defenders), while within-role pools (20–65 players) are too small and unstable, flipping membership under small perturbations
 - **The Scouting Solution:** We shift from rigid clusters to a **nearest-neighbor approach**, answering the operational recruitment question one player at a time: “Given this specific target, who are his closest stylistic matches?”

SCAN FOR THE LIVE DEMO



ACT 4

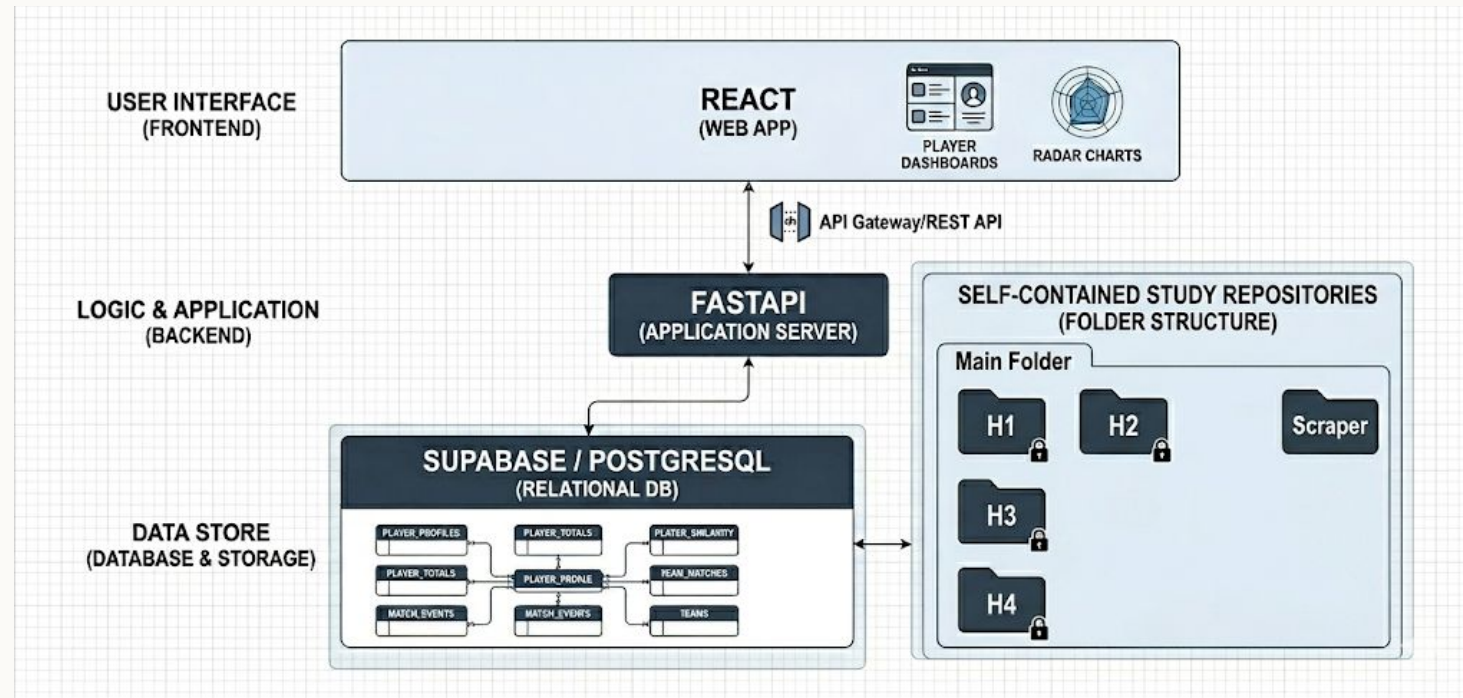
The product

The framework made usable for a scout, then shown live



Architecture, and built to be reproducible

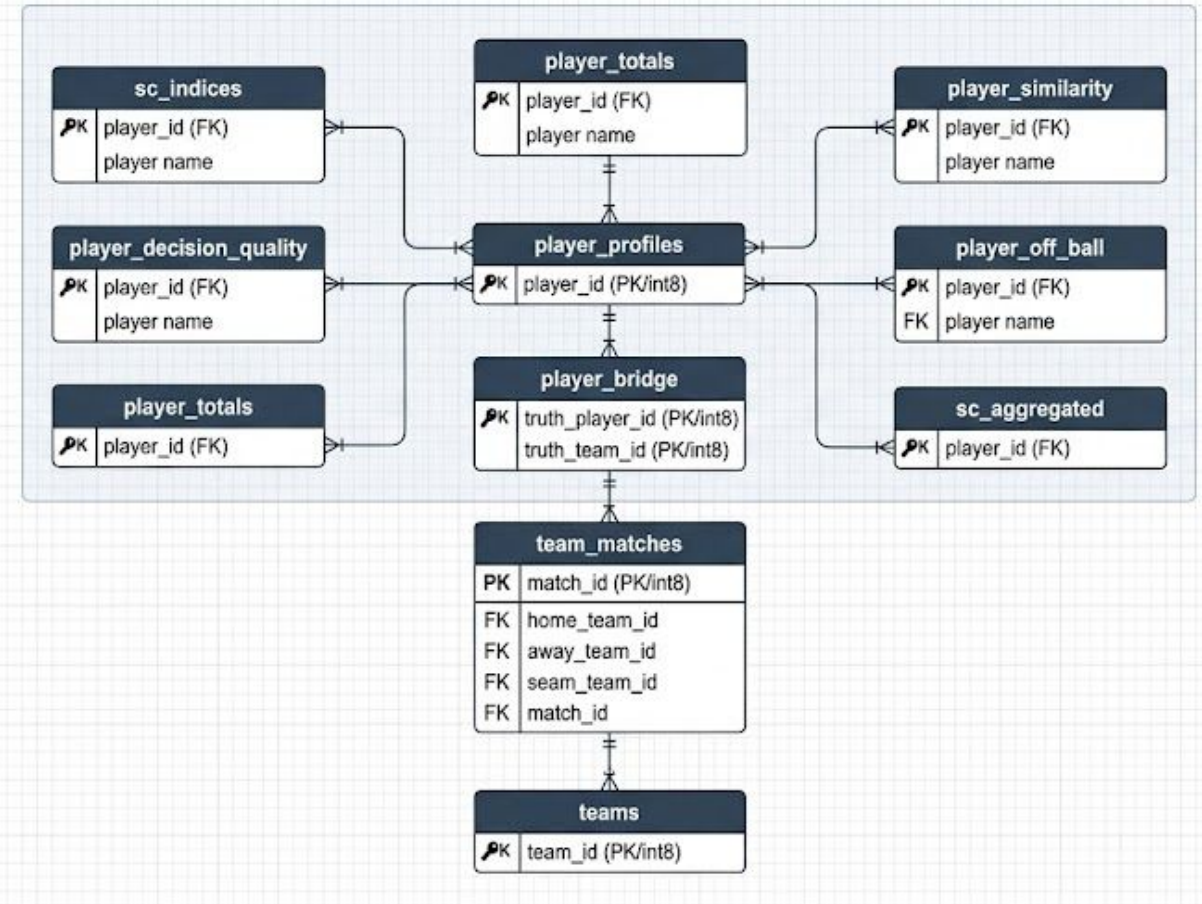
- ▶ The product is a standard **three-tier web application** built with:
 - React frontend
 - FastAPI backend
 - Supabase/PostgreSQL database
- ▶ Built to be **reproducible by design**, each study lives in a self-contained folder with read-only shared player pools to keep all metrics population-aligned



Database Organization



- ▶ The data layer reconciles event-driven spatiotemporal metrics with financial market values and player informations scraped from Transfermarkt
- ▶ A dedicated **data-cleaning** layer maps public StatsBomb player identities directly onto Transfermarkt profiles to eliminate duplication
- ▶ The **relational schema** allows a scout to instantly cross-reference a player's spatial DNA and within-role percentiles against their market valuation





Two types of Research: by Player

- ▶ The player database lets scouts query any of the 272 filtered players to instantly open their respective **unified profile card**
- ▶ There is the possibility to filter by: Role, Macro Role, Age, Nationality, Preferred foot, Value pre-EURO, Value post-EURO, Delta value
- ▶ Clicking any player name opens their full tactical breakdown

Search by Player

All the players with full stats, market values, space-control indices, and value delta. Click any name to open the full profile.

× FILTERS

FILTER BY NATIONAL TEAM

Player Database

1 PLAYERS

PLAYER ▲	ROLE ♦	AGE ♦	NATION ♦	FOOT ♦	PRE € ♦	POST € ♦	Δ VALUE ♦
<u>Lamine Yamal</u>	Right Wing	16	Spain	L	€90,00M	€120,00M	+30.0M (+33%)



Two types of Research: by Attribute

- ▶ Users can find a precise tactical profile by **filtering by specific tactical parameters** rather than names
- ▶ Clicking any player name opens their full tactical breakdown

Search by Attribute

Filter all the players by macro role, tactical role, and contextual space control index ranges. Rankings are sorted alphabetically by player name by default.

Filters

MACRO ROLE

ALLCBFBMIDCAMWIDEFW

PRIMARY ROLE

Right Wing

NATION

Spain

1 players found

ACTIVE FILTERS: MACRO: WIDE ROLE: RIGHT WING NATION: SPAIN UNCAPITALIZED RUN SCORE: 50-100

CLEAR ALL

NAT PLAYER NAME PROG DNGR RECEP GRAV DQ URS AVG MIN

1		Lamine Yamal	84	78	27	41	54	70	58	507'
		RIGHT WING WIDE	PROG	DNGR	RECEP	GRAV	DQ	URS	AVG	MIN

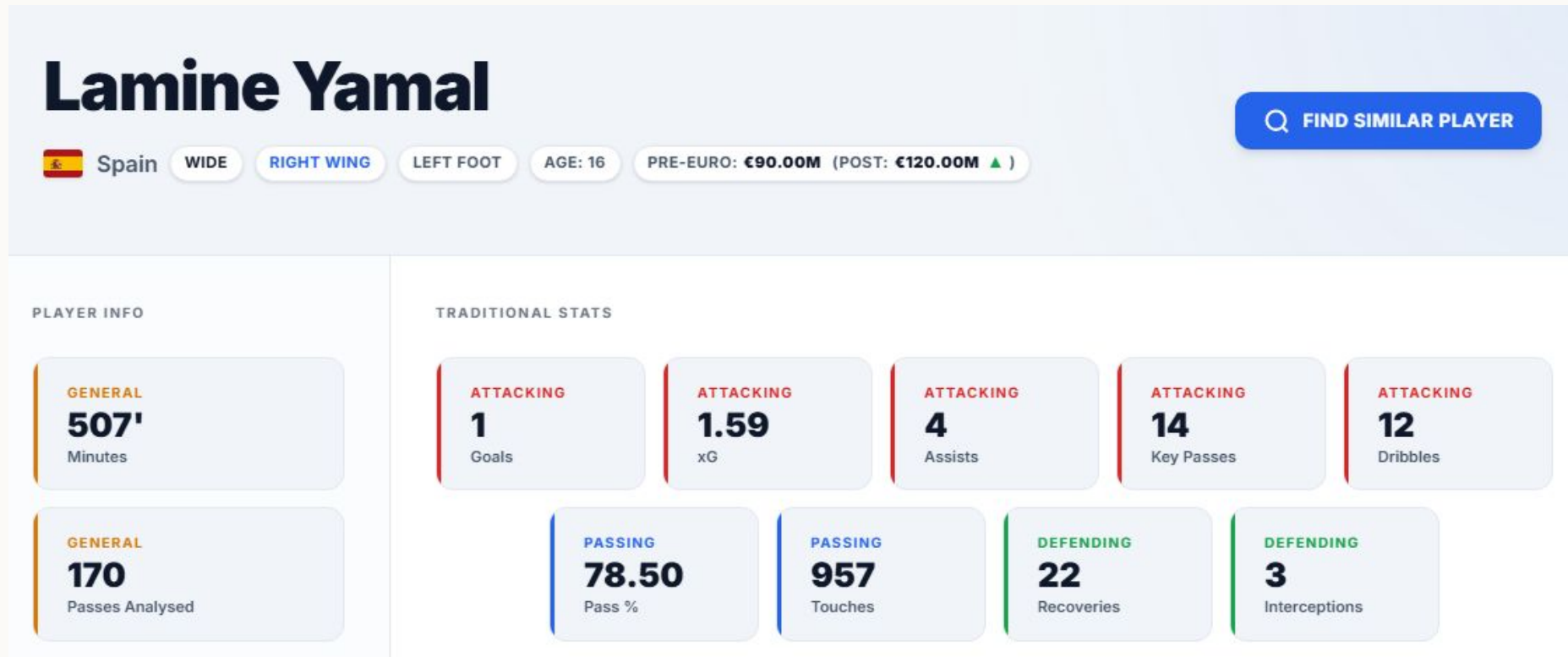
Contextual Football Scouting · Barça Innovation Hub

53



Two types of profile: Player profile

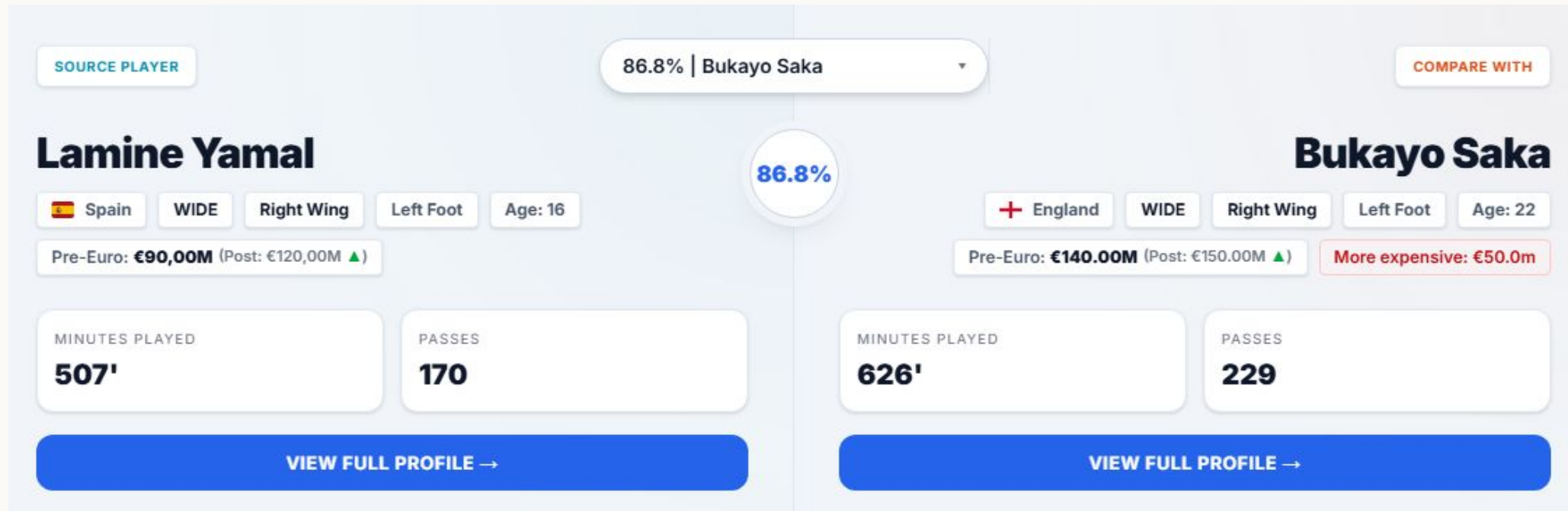
- ▶ In each players dashboard there are the **main statistics** of the player
- ▶ There is the possibility to **find similar players** by clicking the specific button





Two types of profile: Similar players profile

- ▶ The comparison module overlays two players of the **same macro-role**
- ▶ By keeping the similarity within role, the interface **prevents false comparisons** driven by structural positional gaps on the pitch
- ▶ There is a **dedicated drop-down menu** that lets the user select from every comparable player ordered by **similarity score**





Data implementation and Visualization

- ▶ The platform utilizes **tailored radar graphs unique to each hypothesis**, ensuring the polygon's shape immediately reflects the player's style
- ▶ All axes are consistently oriented so that the outside of the radar represents the higher percentile
- ▶ Each radar has the **index score** for that player's stats
- ▶ A dedicated **core stats** block sits directly below the radars
- ▶ There are 3 types of core stats:
 - Raw
 - Per90
 - Percentages

They change based on a toggle located above each radar graph's section

Raw Per 90 Percentages

Player view



Similar Players view





How to read metrics?

- An **interactive glossary** defines the platform's core architectural premises, tracking conventions, and model boundaries for the end user
- It **clarifies that all data** looks at open play only, filters out set pieces, and enforces a mandatory 135-minute floor to reduce noise
- Every single metric and core stat is described and can be consulted in this section

Glossary

Definitions of every metric on the platform. The variables are grouped by hypothesis: Space Control (H1), Decision Quality (H2) and Off-Ball Movement (H3). A few premises are shared by all three and are worth reading once before the individual cards below.

FILTER **ALL** COMMON PREMISES PROGRESSION DANGEROUSNESS RECEPTION GRAVITY DECISION QUALITY OFF-BALL MOVEMENT

Common premises

What the '360 freeze frame' is

For every open-play pass StatsBomb provides 360 data for, we know the (x, y) positions of the visible players at the instant the pass is released: who is behind the ball line, who is in front, where the opponents are. This is what lets us move from "total passes" to "passes in the context they were played in", and it is the shared foundation of all three hypotheses.

Player pool

A player needs at least 135 minutes (about 1.5 matches) to be included: 272 players in total, one row per player. The same pool is used across H1, H2 and H3, so a profile is directly comparable from one hypothesis to the next. Below that threshold the numbers rest on too few events and become noisy.

Macro-roles

Every within-role percentile on the site is computed inside the player's macro-role: CB, FB, MID, CAM, WIDE, FW. A full-back at the 90th percentile is "among the best full-backs", not "among all players". A player's macro-role is the role he actually played most across the tournament, which can differ from his nominal position.

Open play only

All metrics look at open play only. Set pieces (corners, free kicks, throw-ins, kick-offs and goal kicks) are left out, because they are rehearsed situations that follow their own logic and would distort a measure built to read the flow of the game. What each hypothesis then counts underneath differs by design: H1 reads the passes a player attempts, H2 reads the passing decisions he faces, and H3 reads the off-ball positions he takes up while a teammate has the ball.

ACT 5

Future work & Closing

Summary of Findings, Future Work and Final Conclusions

Declared Scope & Future Work

What we declare	Where it goes next
Single tournament (51 matches)	Validation across tournaments and club seasons
No game state in xEPV	A game-state-aware value layer
A single instant, not a film	Integrating tracking data for speed and acceleration
Partial 360 coverage (8 to 12)	Inherent to the public data
Team bias reduced, not removed	Exposure stays the honest boundary
Anonymous teammates (off-ball identities estimated)	Access to official tracking data with full player IDs

Final Verdict

- ▶ One idea, end to end: read a player by what he does with space, by value not volume, within role

WHAT WE BUILT

- ▶ **xPass** and **xEPV**, our own models, on a single public 360 frame (AUC $\approx$ 0.92)
- ▶ **The receiver resolver**, names on the anonymous dots, 93.2% confident
- ▶ **Four studies** on one shared engine, one identical xEPV
- ▶ A **working scouting dashboard** over all 272 players

WHAT IT FOUND

- ▶ Our ranking **re-orders volume**, and it does so sensibly
- ▶ **Off-ball value is measurable** on public data
- ▶ **Cheaper look-alikes** for top-tier targets
- ▶ Decision quality yields a **relative choice-optimality index**, balancing completion probability (**xPass**) against spatial value (**EPV**)
- ▶ The **six axes measure different things**: only **5 players are above the 50th percentile** on all of them (Kovačić, Wirtz, De Bruyne, Rabiot, Škriniar). If the indices they overlapped, far more would

Thank you

Questions & discussion

Matteo Vezzoli & Armando Mio
Contextual Football Scouting · github.com/ArMat-Analytics